

**Mitigating Premature Convergence with Diversity-Preserving Level-Guided
Differential Evolution with Adaptive Mutation**

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Chapter 1

INTRODUCTION

In the domain of computational intelligence, large-scale global optimization (LSGO) presents a critical challenge, as the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, recent advancements such as the LBLDE framework proposed by Qiao et al. [22] have introduced population learning strategies to better balance exploration and exploitation. A significant research gap was identified by Qiao et al. [22], who suggested that their adaptive learning and dynamic population mechanisms could be promising for further enhancement in population-based evolutionary algorithms, specifically DE. This study directly addresses that gap by proposing, designing, and evaluating a new hybrid algorithm, the Dynamic Population Level-based Learning Differential Evolution (DP-LLDE). This research employs an experimental comparative design to test DP-LLDE's performance and scalability against its foundational algorithms DE [26] and LBLDE [22]. On the established CEC'2017 [22] benchmark suite using dimensions of 10, 30, 50, and 100.

Recent research on learning-based population mechanisms, however, suggests that although structured learning increases convergence efficiency, it may also inadvertently decrease particle diversity as particles tend to cluster around high-performing exemplars. In later optimization stages, this loss of diversity may limit exploration, especially in high-dimensional and multimodal landscapes. Therefore, addressing this trade-off between diversity preservation and convergence speed has become a key area of research in evolutionary computation [22].

1.1. Context of the Study

In computational intelligence, optimization in high-dimensional and complex problem spaces remains a significant challenge. The simplicity and robustness of many traditional population-based algorithms, like Differential Evolution (DE), have led to their widespread use in the exploration of nonlinear and multimodal search spaces. However, particularly in large-scale optimization problems, their dependence on crossover and random mutation may restrict guided exploration. This reduces the variety of particles over time and can sometimes lead to premature convergence, in which the population of particles congregates around less-than-ideal locations.

To get around these limitations, hierarchical and adaptive strategies have been proposed. The Level-Based Learning Differential Evolution (LBLDE) framework, developed by Qiao et al. [22], integrates a level-based learning mechanism into the DE algorithm. Particles are divided into fitness-based levels, and lower-level particles pick up knowledge from higher-level exemplars. This allows the algorithm to focus on promising areas while controlling exploration and exploitation. Where it has been shown that LBLDE performs better than classical DE in terms of particle quality and convergence speed.

Despite these improvements, LBLDE may still face challenges or is still prone in maintaining particle diversity later in the optimization process. When many particles converge toward higher-level exemplars, the algorithm may become less variable, which could limit the exploration of other potentially optimal regions in the search space. In order to achieve consistent performance across various large-scale and high-dimensional problem types, it is imperative that this limitation be addressed.

This study focuses on the Diversity-Preserving Level-Guided Learning Differential Evolution (DP-LLDE) algorithm, which extends the level-based learning principle of LBLDE. By dynamically prioritizing fitter particles, DP-LLDE reduces the likelihood that multiple particles will converge to the same location. To preserve diversity and enhance search guidance, it uses sequential relative mutation of particles across specific elite levels. This allows the algorithm to focus on promising particles while at the same time exploring other areas in the solution space.

Moreover, ideas from computational models of attention mechanisms have been incorporated into DP-LLDE. Such mechanisms are efficient in that they reduce redundancy in evaluation and instead concentrate more on individuals with higher fitness evaluations. Given that deep learning attention gives priority to important features or representations, this implies a conceptual relationship between level-based evolutionary optimization and attention-based learning.

In this study, optimization problems are considered as benchmarks to evaluate the effectiveness of DP-LLDE, particularly in terms of efficiency and quality of particles. This study illustrates how attention-inspired focus, adaptation, and learning can help to improve the performance of evolutionary algorithms in high-dimensional problem spaces.

1.2. Research Gap

A limitation shared by population-based algorithms is highlighted in the main paper on Level-Based Learning Differential Evolution (LBLDE) by Qiao et al. [22]: even though the level-based learning mechanism enhances convergence, the particles in some benchmark problems still fall short of the global optimum. This suggests that more

literature is further required to improve the direction and efficacy of level-based learning in the field of Differential Evolution.

Although LBLDE uses higher-level exemplars to guide lower-level particles and groups particles into fitness-based levels, this process may reduce population diversity, a specific research gap identified by Qiao et al. [22]. As more particles converge toward the higher-level exemplars, the algorithm might focus on fewer regions of the search space, which would limit exploration. In high-dimensional optimization problems, where preserving particle variability is crucial to avoiding premature convergence, this is especially difficult.

This gap provides an opportunity to explore approaches that maintain particle diversity while still taking advantage of hierarchical learning. The Diversity-Preserving Level-Guided Learning Differential Evolution (DP-LLDE) framework tackles this by giving higher priority to fitter particles and using sequential relative mutation across selected elite levels. By directing particles towards potential areas of the search space while preventing them from crowding too closely, our approach promotes a more balanced combination of exploration and exploitation.

In this work, benchmark optimization problems will be used to assess the effectiveness of DP-LLDE, examining the impact of level-based guidance and diversity-preserving techniques on particle movement and search performance. Additionally, attention-inspired focus reduces superfluous calculations and improves overall optimization outcomes, focusing computing effort on the most promising particles.

1.3. Research Questions

The aforementioned objectives were formulated to address the following research questions:

1. How does the proposed Diversity-Preserving Level-Guided Learning Differential Evolution (DP-LLDE) algorithm perform in comparison to its foundational algorithm (LBLDE) across benchmark functions?
2. Can DP-LLDE's combination of level-based learning and sequential relative mutation guide particles to achieve superior or more consistent solution quality on the CEC 2017 Large-Scale Global Optimization Benchmark functions across different dimensions (10, 30, 50, and 100)?
3. When applying DP-LLDE to high-dimensional optimization problems, what are the trade-offs between computational cost and convergence speed, and how does the algorithm scale as particles explore increasingly large solution spaces, particularly in 50- and 100-dimensional problems?

1.4. Research Objectives

The main goal of this study is to assess the effectiveness of the Diversity-Preserving Level-Guided Learning Differential Evolution (DP-LLDE) in solving large-scale optimization problems. Specifically, this study aims to.

1. design and develop the DP-LLDE algorithm, including sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. compare the performance of DP-LLDE with the base algorithm LBLDE [22] using the CEC 2017 Large-Scale Global Optimization Benchmark functions.

3. examine the DP-LLDE particle behavior, convergence rate, and solution quality, with special focus to how the results for the different issue dimensions (10, 30, 50, and 100) are affected by diversity-preserving and level-based guiding strategies.

1.5. Significance of the Study

The objective of this study is to empirically address the future work proposed by Qiao et al. [22] in order to advance the field of evolutionary computation. regarding the application of their level-based learning strategy to Differential Evolution. Through evaluating DP-LLDE against a suite of benchmark functions, this study may provide a more effective hybrid approach for large-scale optimization, addressing issues where particle solutions converge prematurely or become trapped in false local optima. The findings may offer researchers a reliable and efficient tool for obtaining solutions "as near the global optima as possible," while serving as a reference point for performance on standard benchmarks such as the CEC 2017 Large-Scale Global Optimization functions across 10, 30, 50, and 100 dimensions.

1.6. Scope and Limitations

The Diversity Preserving Level-Based Learning Differential Evolution (DP-LLDE) algorithm's design, implementation, and assessment will be the main topics of this work. In order to empirically address the future work mentioned by Qiao et al. [22], this study will utilize an experimental comparative design. who suggested investigating the application of their level-based learning strategy to Differential Evolution (DE). The "particles" in the population will represent candidate solution vectors in a D-dimensional continuous space, and their "fitness" will be determined by the objective function values of the benchmark problems.

The experimental evaluation will use the CEC 2017 Large-Scale Global Optimization Benchmark functions, which are specifically designed to test high-dimensional optimization performance. The performance of DP-LLDE will be compared against its foundational algorithm (LBLDE [22]). The algorithm's scalability will be explicitly tested on problems with 10, 30, 50, and 100 dimensions.

The study will be limited by computational resources, which may constrain the maximum number of function evaluations used as the stopping criterion. Furthermore, the evaluation will only be conducted on synthetic benchmark functions, and the findings could not apply directly to black-box optimization problems in the actual world.

Chapter 2

REVIEW OF RELATED LITERATURE & STUDIES

This section shall discuss thoroughly how optimizers and attention mechanisms share underlying principles of adaptive learning and selective focus upon testing with complex problem spaces. With evolutionary and swarm-based algorithms, DP-LLDE aims to optimize by dynamically prioritizing fitter candidate solutions while maintaining variability, diversity, and avoiding the probability of cloning elite solutions for most particles. Through sequential learning through relative mutation of learning particles in selected elite levels. Including deep learning computational focus utilizing attention mechanisms. These frameworks examine attention from the perspective of optimization – how performances are presented and how it highlights fundamental strategies developed for optimizing neural networks and it's fit by conceptually parallel adaptive mechanisms in swarm intelligence.

2.1. Evolutionary Algorithms and Global Optimization

Natural selection and Darwinian evolution serve as the inspiration for a large class of population-based, stochastic optimization methods known as evolutionary algorithms (EAs). Their primary advantage is their capacity to search large, nonlinear, and multimodal spaces without the need for gradient information. Through certain selection, crossover, and mutation operators, EAs keep a population of potential solutions that change over the course of multiple generations [1]. Each particle stands for a possible solution that is assessed using a fitness function that gauges how well it meets the goal of the problem. The population progressively approaches an ideal or nearly ideal solution by iteratively favoring fitter individuals.

The Genetic Algorithm (GA), one of the first and most significant EAs, mimics biological reproduction by using crossover and mutation to preserve diversity while selection pressure encourages individuals of higher quality. There are some successful applications that were cited in fields like neural-network weight optimization, antenna-array design, and scheduling have been made possible by GA's adaptability [12]. Particle Swarm Optimization (PSO), another fundamental method, was first presented by Kennedy and Eberhart in 1995 and models the collective behavior of fish schools and bird flocks. The global best (gbest) and personal best (pbest) positions are used by each particle to modify its velocity. According to Al-Tashi et al. [1], PSO is a cornerstone of swarm-intelligence research due to its ease of use, quick convergence, and low number of parameters. However, when populations become less diverse and congregate in less-than-ideal areas, GA and PSO may both experience premature convergence.

2.2. DE

Building on these literature, Differential Evolution (DE), a closely related but separate technique, uses an entirely different approach which is vector-based operations to approach optimization [23] [26]. By altering current candidate solutions with scaled differences of randomly chosen population members, it creates new ones and enables the search process to self-adapt between exploration and exploitation [20]. Due to its efficacy in continuous optimization, DE is widely used in real-valued engineering problems, benchmark testing, and parameter tuning [17] [27]. Even though DE is robust, if mutation step sizes are not properly adjusted, performance may suffer in high-dimensional spaces. Therefore, in the recent developments there have been concentrated advancements on adaptive and hybrid frameworks that dynamically balance exploration and exploitation,

such as the Improved Fitness-Dependent Optimizer (IFDO) [20] and the Adaptive Meta Optimizer (ATMO) [14]. In order to improve and enhance its guidance, adaptability, and convergence in challenging global optimization problems especially in very complex spaces, these advancements lay the theoretical groundwork for extending DE with higher-level learning strategies, like the Level-Learning mechanism.

An illustrative case of the DE technique has been shown by Storn and Price (1997) [26], who implemented DE in several nonlinear and multimodal optimization problems. Experimental results confirmed the ability of DE to reach near global optima with fewer control parameters compared to the Genetic Algorithm. Still, the study showed the degradation of the DE technique with the increase in problem size unless the step sizes involved in the mutation operator are well managed. Further empirical studies have supported these findings. Qin et al., (2008) [23] implemented DE on complex global numerical optimization problems and discovered its effectiveness through adaptive control of control parameters during mutation and crossover operations. This led to implementing adaptive DE variants like JADE and its better performance over classical DE on high-dimensional functions. Yet these advances in DE algorithms notwithstanding, in more recent publications it was shown that standard and adapted forms of DE algorithms may still be subject to early convergence and poor guidance in optimization of large-scale problems. The addition of a learning method of level type to DE was performed in [22], Qiao et al., 2022, and was shown to yield convergence in less time and solutions of higher quality than standard DE but also to result in reduced diversification of solutions in later stages of evolution.

2.3. LLSO

Another yet significant development in swarm intelligence that supports this literature, is the Level-Based Learning Swarm Optimizer (LLSO), put forth by Chen et al. [7], adds a hierarchical learning mechanism to enhance exploration and exploitation in large-scale optimization. LLSO from this sense, simulates a teacher-student environment in which candidate solutions, referred to as particles, are ranked and categorized into levels based on fitness, in contrast to conventional algorithms like Particle Swarm Optimization (PSO) or Differential Evolution (DE), which primarily rely on velocity or mutation operations.

While top-level particles continue to improve themselves through limited exploration, lower-level particles learn from those in higher levels as a result of balancing between exploration and exploitation in early and later stages of evolution. Then, through the random selection of exemplars from higher levels or tiers, this structure preserves diversity and learning efficiency, enabling the swarm to avoid premature convergence by letting solutions mimic the exemplars excessively. A constant feedback loop between exploration and improvement is produced by reassigning particles according to updated fitness at each iteration.

The computational cost of LLSO scales with problem dimensionality, following the empirical relation:

$$3000 \times \text{dimensions} = \text{no. of evaluations} \quad (1)$$

This increases runtime but guarantees sufficient search for high-dimensional problems [7]. However, experiments demonstrated that LLSO performed better than traditional algorithms like PSO, Genetic Algorithm (GA), and DE across benchmark

functions, especially in complex search landscapes where traditional methods frequently stall. The hierarchical structure of LLSO is consistent with the general movement toward optimization that is adaptive and self-learning. The distinguished idea of balancing exploration and exploitation through adaptive weighting or dynamic strategy integration is shared by related algorithms like the Adaptive Meta Optimizer (ATMO) [12] and the Improved Fitness-Dependent Optimizer (IFDO) [20] in order to accomplish a focused and ideal learning of the search space.

In the study on the performance assessment of the Level-Based Learning Swarm Optimizer algorithm using large-scale global function optimization problems, the authors Chen et al. [7] explored the performance of LLSO on high-dimensional multifunction problems regarding the speed of convergence and the quality of the resultant solutions compared to classical paradigms like classical Particle Swarm Optimizers (PSO), Genetic Algorithms (GA), and classical Differential Evolution (DE). For the authors, the success of the modern hierarchical approach was due to the capacity for weaker members within the system to benefit in terms of fitness while allowing the system to avoid the local optima problem more successfully than before. Although, the robustness of LLSO, Chen et al. [7] disclosed that LLSO presented remarkable limitations that encouraged improvements to the LLSO. It was claimed that while LLSO performs well, it becomes computationally expensive when rendered to solve large problems. Additionally, even though the random choice of the particle's position from the set of exemplars promotes diversity, yet an increased period of learning from a particle of a relatively lower level may result in similar accumulation in latter evolutionary stages. It was to address the shortcomings that DLLSO

came into being. Besides, it was the issues that were resolved in more refined diversification of the level-based evolution.

2.4. DLLSO

To overcome the Level-Based Learning Swarm Optimizer's (LLSO) high computational cost and constrained scalability in large-scale optimization problems, the Dynamic Level-Based Learning Swarm Optimizer (DLLSO) was presented as an improvement [7]. Another is that, DLLSO introduces dynamic adaptation, which enables the number of levels and learning intensity to automatically adjust based on problem complexity and search progress, whereas LLSO uses a fixed number of learning levels.

DLLSO has the capacity and is able to better balance exploration and exploitation thanks to this adaptive mechanism. Wherein in the early stages, additional levels are added to encourage exploration in multiple ways; and as convergence improves, the levels are reduced gradually by increasing the search space around promising areas. This ensures robustness and resistance against getting trapped in a local optima while ensuring convergence is achieved in a quicker and proper manner that does not reduce its chances to reach the global optimum.

Research has shown that DLLSO outperforms LLSO and traditional evolutionary algorithms like PSO, GA, and DE in terms of convergence speed, scalability, and solution quality. Therefore, based on this self-adjustment and adaptability capacity, it is then clear that it will perform better in a high-dimensional and/or dynamic environment in which other static optimization techniques may fail or become stuck. According to previous

studies [7], DLLSO achieved higher accuracy with fewer evaluations, suggesting increased computational efficiency without compromising diversity.

The self-adaptive and knowledge-driven optimization trend in evolutionary computation is reflected in DLLSO. Similar to fitness-dependent models [20] and adaptive swarm strategies [1], DLLSO uses a learning hierarchy that changes in response to feedback from the environment while searching.

Importantly, DLLSO lays out the fundamental theoretical underpinnings of this research, which is the idea of level-based learning as an organized and flexible method of information transfer. Because of its hierarchical mechanism, collective search efficiency is increased as lower-performing individuals can learn from higher-performing exemplars. The proposed Level-Learning Differential Evolution (LLDE), which incorporates the same adaptive learning process into the mutation and selection stages of Differential Evolution, is directly inspired by this fitness-guided hierarchy principle. Therefore, it is essential to comprehend DLLSO since it outlines how guided learning and dynamic hierarchy can enhance optimization performance across various algorithmic frameworks.

However, this solution mostly allows the cloning of elite solutions (i.e., higher level particles) from lower class arrive at seemingly close solutions that reduce variability and increase systemic bias. The mechanism behind the nature of DLLSO relies on how its computed mutant vector from low-level to high-level has a predetermined chance where it overrides the current vector to take that solution. This is one that prevents most particles from arriving at similar solutions. But learning particles are always coming from the lowest level and the particles to learn from reside in the highest level. While probability creates a

chance for the initial vector from mutating to the mutant vector, this fails to explore other levels of the hierarchy.

2.5. Comparative Evolutionary Optimization

Recent trends in optimization studies indicate that the scope of machine learning fine-tuning approaches and evolutionary optimization algorithms is becoming similar in their emphasis on the need for improved performance, parameter efficiency, and flexibility. In this regard, the recent optimization approaches incorporate dynamic adjustment and feedback mechanisms in order to improve convergence and generalization, while the traditional optimization algorithms like PSO, GA, and DE rely on static parameters and population-based search [1].

Analogous optimization dynamics are observed with parallel advancements in fine-tuning studies as well. In a study [20], the researchers were able to prove that the integration of Bayesian optimization with efficient fine-tuning significantly enhances the learning process while reducing the computational cost, as observed with the use of DLLSO, an adaptive optimizer that changes its level structure based on the complexity of the problem. In a similar manner, another study [3] proposed selective adaptation in parameter-efficient tuning, which maintains the accuracy while keeping the computational cost in check. Like fitness-based learning in evolutionary computation, Lalor et al. [16] introduced Crowd-Informed Fine-Tuning (CIFT), in which item-level difficulty directs model updates. Both depend on the hierarchical transfer of knowledge, in which weaker agents—such as the teacher-student structure in LLSO—are informed by stronger exemplars.

Adaptive hybridization is further emphasized in recent studies. The Improved Fitness-Dependent Optimizer (IFDO) proposed by Muhammad et al. in [20] was observed to converge better and be more stable compared to the traditional PSO and DE algorithms using the concept of dynamic weighting. For faster convergence, Gallo et al. [12] presented an Adaptive Meta Optimizer, abbreviated as ATMO, which incorporates learning methods such as Adam and SGD with a meta-learning approach. Similarly, Keskar and Socher [14] presented a learning approach called SWATS, where adaptive learning methods are replaced with SGD for better generalization. These advancements in learning methods, as discussed above, indicate the importance of adaptive learning for achieving faster convergence, accuracy, and scalability for swarm intelligence as well as fine-tuning.

2.6. Attention and Adaptive Optimization

Instead of treating information equally, attention mechanisms in computational models allow a system to focus more on the most relevant information in the input data [2] [6]. Attention mechanisms allow learning systems to better manage computational resources by providing more weight to important features or representations in the input data. The concept of attention mechanisms is analogous to adaptive optimization algorithms that change search strategies based on the complexity of a problem and the quality of potential solutions [17] [27].

This principle closely resembles mechanisms such as those found in the Dynamic Level-Based Learning Swarm Optimizer (DLLSO) in the context of evolutionary and swarm-based optimization. DLLSO focuses its computational "focus" on higher-performing solution levels, dynamically adjusting its structure to highlight promising areas of the search space, much like attention models prioritize important features during

learning. Reducing redundant computation while increasing effort in areas with greater potential is the aim of adaptive learning, which is embodied by both frameworks.

Furthermore, several studies [5] [11] have demonstrated that using a linear or fine-grained attention mechanism is more efficient, especially if information is prioritized according to context. DLLSO achieves optimization efficiency, similar to how fitter agents are prioritized, by adjusting its hierarchy as convergence is reached. This demonstrates how attention mechanisms have permeated beyond deep learning, affecting the creation of more general intelligent optimization techniques.

2.7. Attention-Guided and Reinforced Learning

Attention and reinforcement learning theories have emphasized the role that adaptive systems use in attending to the most relevant information in order to solve problems effectively [21]. In a study by Daniel et al. [9], it has been shown that “just as optimization methods bias search in promising regions of the space, reinforcement learning in multidimensional spaces relies on attention mechanisms to filter informative cues and expedite learning.”

While Khan et al. [15] found that attention is crucial for striking a balance between exploration and exploitation in contemporary machine learning, Deshpande et al. [10] demonstrated that scaling models with attention-based focus increases training efficiency in artificial systems. Similar to adaptive control parameters in evolutionary algorithms, Lalor et al. [16] discovered that soft attention weighting improves generalization.

Collectively, these studies suggest that the function of attention is as a selective learning mechanism, channeling cognitive effort into useful information. This perspective

is consistent with the integration of attention-based focus into optimization models like the proposed LLDE, as with the idea behind Level Learning, in which weaker solutions are based on stronger ones.

2.8. Conceptual Framework

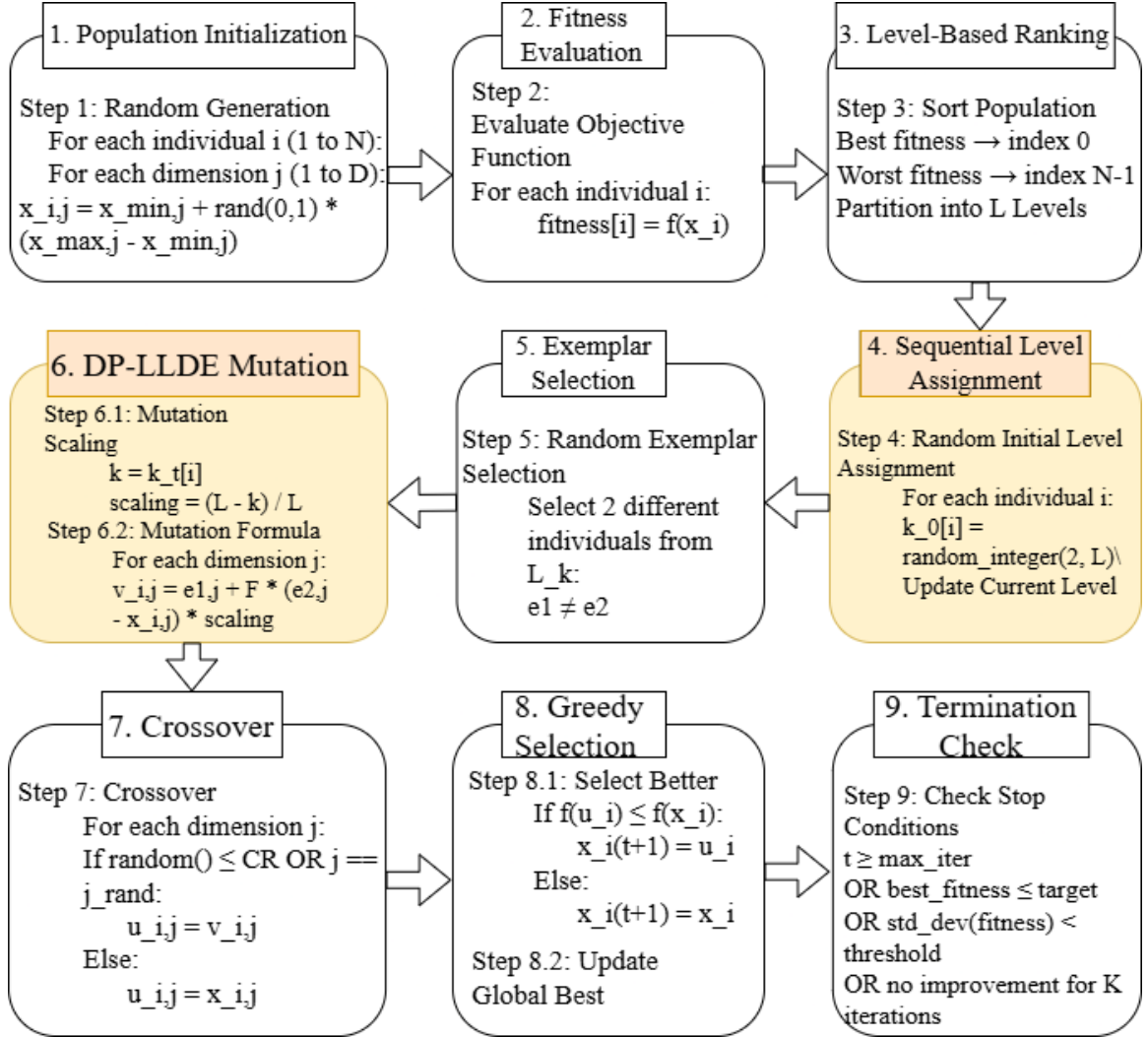


Figure 2.1. Proposed Algorithm Framework

Figure 2.1 shows the overall workflow of the proposed algorithm illustrating the main evolutionary steps from initialization to termination. With two additional steps (Sequential Level Selection, DP-LLDE Mutation) utilized in this study.

2.9. Theoretical Framework

This subsection defines the theoretical integration of Differential Evolution (DE) and the Level-Based Learning (Learning) strategy mechanism derived from Level-Based Learning Swarm Optimizer (LLSO) from related studies [7]. This hypothesizes that incorporating these two distinct algorithms can enhance convergence and global optimization to that of the results attained by Chen et al. [7]. These two points: 1) Applies structured, and hierarchical ranking based on fitness of a possible solution; 2) Applies the mathematical computation of how DE improves its fit by unstructured, neighboring vectors.

2.9.1. Differential Evolution Formula

This evolution algorithm (EA) is a population-based stochastic optimization designed for continuity in certain parameter spaces. Which contains a set of candidate solutions, or in other words, a list of students in a syllogistic analogy [7].

This algorithm exhibits the concept of evolution from a set of candidate solutions $\{x_1, x_2, \dots, x_i\}$ such that these x_i is possible candidate solution where $x_i \in \mathbb{R}^d$ represented in a D-dimensional space [7] [8] [23]. These solutions operate under three (3) main phases: mutation, crossover, and selection:

1) Mutation:

Each x_i is subject to three distinct individuals x_{r1}, x_{r2}, x_{r3} that are randomly selected from the population and are calculated from the formula:

$$v_i = x_{r1} + F(x_{r2} - x_{r3}) \quad (2)$$

Where $F \in (0,1]$ is a scaling factor what controls the strength of the mutation, $(x_{r2} - x_{r3})$ represents the direction that DE must go that might lead to improvement. Intuitively adding x_{r1} to that difference creates a new candidate that moves in that calculated direction [13] [22] [26] [29].

2) Crossover:

As v_i now becomes the mutant vector, meaning a possible solution that was learned from three points – it mixes with the original vector x_i to form a trial vector u . For each dimension j :

$$u_j = \begin{cases} v_j, & \text{if } \text{rand}() < CR \\ x_{ij}, & \text{otherwise} \end{cases} \quad (3)$$

Where CR is the crossover rate (a value between 0 and 1), this is a measured proportion of how much the calculated mutant vector in a D-dimensional space will replace a percentage of the D-dimensional space of the original vector. For instance, if $CR = 0.9$ this means that 90% of the value in x_i is going to be replaced by the mutant. To avoid bias in training, $\text{rand}()$ generates a uniformly distributed number between 0 and 1 for CR to attain a given percentage on what retains and changes [13] [23] [26] [29].

3) Selection:

Intuitively, rather than blindly retaining the calculated trial vector, its fitness still must be evaluated from:

$$x_i^{(t+1)} = \begin{cases} u, & \text{if } f(u) < f(x_i) \\ x_i, & \text{otherwise} \end{cases} \quad (4)$$

Where if u is better, in other words lower cost, higher fitness, it would be kept for the next generation [13] [23] [26] [29].

2.9.2. Level-Based Learning Formula

This optimization inhibits a population of candidate solutions where each solution has a fitness value, that measures how well it can solve the problem [7]. These populations are ranked by their fitness and divided into levels $\{L_1, L_2, \dots, L_n\}$ where the top one mainly consists of the best exemplars, and the second are the second exemplars. The lowest level are individuals with the lowest performance and LL attempts to address this by training lower-level points to higher level points to improve their fitness value and ultimately to approach the best weights for problem complexities [7]. This is represented through the formula:

$$x'_i = x_i + \alpha_1(x_p - x_i) + \alpha_2(x_q - x_i) + \epsilon \quad (5)$$

Where two higher-level exemplars (x_p, x_q) are chosen from the condition of $x_p \in L_m, x_q \in L_n$ that should be $m, n < k$ meaning these exemplars must belong to higher levels compared to the lowest one. This equation particularly this region: $\alpha_1(x_p - x_i) + \alpha_2(x_q - x_i)$ calculates the direction of the learner towards the two chosen exemplars with α_1, α_2 being small positive numbers between 0 and 1 that controls how much the learner moves towards each exemplar. With ϵ being a small random perturbation that is meant to prevent calculated individuals from being identical to each other.

2.9.3. Integration of Diversity Preserving LLDE

Having grounded mathematical foundations underlying both DE and LL strategies, it is now possible to synthesize and formulate a combined theoretical formula of the two main concepts. The proposed Diversity-Preserving Level-Learning Differential Evolution (DP-LLDE) algorithm now applies hierarchical learning structures from LL into mutation

stages of DE. However, low level particles undergo a randomized probability of selecting which level to select from. Where the rule is:

$$k_o \sim \{2, 3, \dots, L\} \quad (7)$$

As individuals do not immediately learn from the elite level, they instead undergo a chance of learning sequentially from preceding levels. The concept of learning randomly sequentially was coined by Chen et al., [7] with considerations on LL strategies that learning individuals in different levels have diverse potential in exploration and exploitation – which subsequently evolve the swarm (i.e., individuals). Thus, the researchers exemplified a new exemplar selection method that is incorporated into this learning strategy, which must “first randomly select two different higher levels” to promote the likelihood of diversity. Given that, predominantly restricting LL strategies to immediately learn from highest one guarantees high quality in the difference vector, however, limits the diversity to calculate the next mutation magnitude to move in the search space [22]. As the next N generation as individuals have minimal distance from each other.

Additionally, after learning whether the mutated vector becomes the new vector or not, based on a set iteration, the upward sequential rule is presented:

$$k_t = \max(1, k_o - t) \quad (8)$$

Where it guarantees that randomized learning from levels always proceeds strictly downward. This formulation ensures that individuals initially assigned to higher (lower quality) levels gradually transition toward the elite level ($k=1$) over successive generations [22]. The use of the lower bound prevents invalid level assignments and guarantees that once the elite level is reached, no further downward transitions occur.

During these mutations, instead of choosing neighboring vectors to improve fitness – the strategy instead selectively samples two higher-level exemplars in the population hierarchy from the selected level based on its iteration. The synthesized mutation rule then becomes:

$$v_i = e_1 + F(x_{pbest} - x_i) + F(e_2 - x_i) \cdot \frac{L - k}{L} \quad (9)$$

The now normalized factor $\frac{L-k}{L}$ appropriately scales the differential variation, thereby modulating mutation probability based on relative level. Qiao et al., [22] noted its shortcomings found in equation (5) from introducing a random perturbation ϵ that potentially prevents elite cloning where it will additionally converge to one specific region of the decision space. However, the said vector given its randomized nature, its influence on the population decreases gradually. The researchers additionally pointed out that DE must have the exploration ability in the early stages of learning in N generations and the capacity to exploit or refine the current learning from previous generations in the later stages of differential evolution in a level-based setting.

Where e_1, e_2 are the exemplars drawn from those in the selected level than x_i . This difference equation ensures that the direction of the differential variation from $(e_2 - x_i)$ reflects knowledge-guided improvement rather than supporting random exploration. Where L then represents the total number of levels, while k is the level at which the particle is learning from. This guarantees that this added calculation normalizes mutation pressure across the number of levels it attempts to learn. While also guaranteeing monotonic decay of mutation as learning quality improves and decouples mutation behavior from absolute level indices. Additionally, crossover and selection processes as discussed previously from DE, will remain identical to retain DE's nature of exploration (randomized percentage in

taking a computed mutant value) while also introducing fitness-based guidance. That pulls lower-level individuals toward higher-level exemplars of the search space.

2.10. Synthesis

The literature reviewed indicates a major evolutionary shift in the approach to global optimization algorithms from simple and static population-based heuristics to adaptive and learning-based methods that heavily rely on guidance and hierarchy in complex search environments. While traditional evolutionary and swarm algorithms, including Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Differential Evolution (DE), afford solid and reliable frameworks for developing global optimization algorithms because of their nature as population-based and stochastic processes, these and similar algorithms always exhibit critical inefficiencies, including unnatural convergence and lack of guidance and hierarchy in the search for critical solutions. In attempting to address some of these fundamental shortcomings and deficiencies, adaptive variants of these algorithms, including JADE, the Improved Fitness-Dependent Optimizer (IFDO), and the Adaptive Meta Optimizer (ATMO), have offered useful solutions for improving the nature and state of convergence. Despite these useful contributions, these algorithms and similar ones heavily rely on unstructured population interactions and lack adequate mechanisms for information transfer. In contrast, the level-based learning approaches, such as the Level-Based Learning Swarm Optimizer (LLSO) and its dynamic version DLLSO, introduce explicit hierarchical structures that organize the population according to fitness, allowing weaker individuals to learn from stronger exemplars. This structured knowledge transfer has been shown to dramatically improve convergence speed and solution quality, in particular for challenging large-scale global optimization tasks. However, empirical

results also point out intrinsic trade-offs, as the elite-centric learning increases computational cost and may shrink the diversity of the population, causing solution clustering and higher risk of bias in later stages of evolution. Parallel developments of attention mechanisms and adaptive fine-tuning strategies in machine learning further highlight the importance of selective focus, adaptive resource allocation, and feedback-driven learning, which underlines a strong conceptual overlap between attention-guided learning and hierarchical optimization strategies in evolutionary computation.

Currently available level-based optimizers and optimizers inspired by attention arguably demonstrate weaknesses in balancing exploration and exploitation, mainly related to cloning solutions, limited levels interaction, and limited diversity preservation. These shortcomings point to the actual need to develop an optimization framework capable of maintaining diversity in possible solution particles with probabilistic flow across hierarchical levels. This actual need aligns directly with the purpose of proposing Diversity-Preserving Level-Learning Differential Evolution (DP-LLDE) as an optimizer capable of properly integrating the structured learning guidance with the hierarchical learned process and irrespective of diversity preservation

Chapter 3

METHODOLOGY

This chapter shall describe and define the research methodology used for the design, implementation, and evaluation of the proposed Diversity-Preserving Level-Based Learning Differential Evolution algorithm.

3.1. Research Design

The study employs an experimental and comparative research design that focuses on algorithmic advancement in large-scale optimization. The rationale for the study was based on the limitations and gaps observed in the original LLDE, as applied in the resolution of high-dimensional issues.

In the LLDE algorithm, the population is divided into different levels based on their fitness ranking. The lower the level of the population, the higher the level of learning. This process not only improves the rate of convergence of the algorithm but also, as suggested by previous studies and preliminary analysis, the population tends to become similar at higher levels of evolution. This is the major reason for the problem of premature convergence of the algorithm, as it is unable to move out of the suboptimal region of the search space and avoid local optima, especially in the case of a high-dimensional and complex optimization problem.

The original developers of the LLDE algorithm had suggested that diversity is an area that needs improvement and that the adaptive process and diversity-enhancing strategies need to be studied in the future. Given this research gap, the current study presents a diversity-preserving variation of LLDE that tries to reduce similarity accumulation at higher levels.

Rather than increasing the number of levels or learning structures, the method that is being suggested works towards limiting the impact of mutant vectors on the evolutionary process. The method that is being proposed works towards maintaining exploration ability while preserving the benefits of convergence of LLDE by limiting the impact of substantially similar modified solutions. This design choice clearly solves the diversity degradation concern mentioned in previous work while also aligning with LLDE's stated future research priorities.

3.2. Theoretical Foundation

The proposed method builds upon two established optimization paradigms:

1. **Differential Evolution** – a population-based evolutionary algorithm that evolves potential solutions through mutation, crossover, and selection.
2. **Level-Based Learning Strategy** – Hierarchical learning approach in which candidate solutions are ranked based on fitness and weaker individuals learn from stronger exemplars.

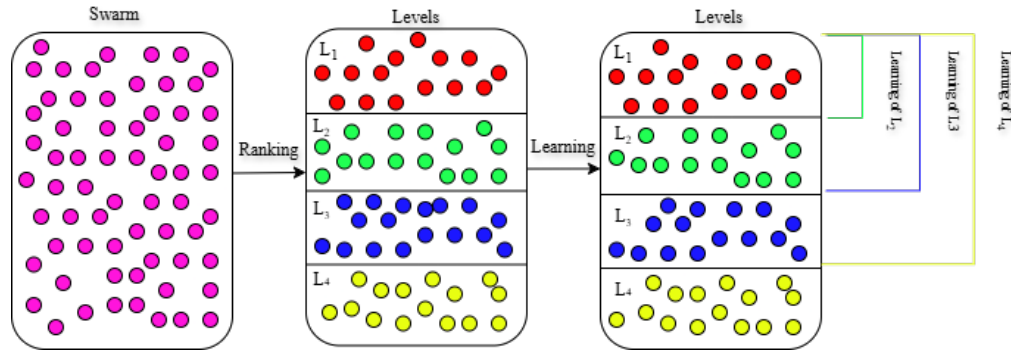


Figure 3.1. Level-Based Learning Strategy

The DP-LLDE architecture retains the essential DE operators while incorporating level-based learning and extra methods to prevent premature loss of population diversity.

3.3. Pseudocode

Algorithm 1: LBLDE (Base Algorithm)

Input: NP (population size), NL (the number of levels),
 LS (number of individuals in each level), $MaxFES$ (maximal number
of the function evaluations)
Output: optimal solution X and its fitness $f(X)$

- 1 Set $FES = 0$, $G = 0$, $\mu_F = 0.5$, $\mu_{CR} = 0.35$, $c = 0.1$;
- 2 Create a random initial population P and evaluate each individual X in
 P ;
- 3 $FES = FES + NP$;
- 4 **while** $FES \leq MaxFES$ **do**
- 5 $G = G + 1$;
- 6 Sort P by the fitness in ascending order and partition it into NL
levels equally;
- 7 **for** $i = 1$ **to** NL **do**
- 8 **for** $j = 1$ **to** LS **do**
- 9 Generate $CR_{i,j} = rand_n(\mu_{CR}, 0.1)$ and $F_{i,j} = rand(\mu_F, 0.1)$;
- 10 **if** $i == 1$ **then**
- 11 $p_i = 0.05$;
- 12 **else**
- 13 $p_i = \frac{(i-1) \times LS \times 100\%}{NP}$;
- 14 Randomly select an individual from top p_i individuals as
exemplar, $X_{best}^{p_i}$;
- 15 Randomly select two individuals, $X_{r_1,j}$ from top p_{i-1} and
 $X_{r_2,j}$ from top p_i individuals;
- 16 Generate the mutation vector
 $V_{i,j} = X_{i,j} + F_{i,j}(X_{best,i}^p - X_{i,j}) + F_{i,j}(X_{r_1,i} - X_{r_2,i})$;
- 17 Generate the trial vector $U_{i,j}$ by implementing crossover
operation between $X_{i,j}$ and $V_{i,j}$ and evaluate $U_{i,j}$;
- 18 $FES = FES + 1$;
- 19 **if** $f(X_{i,j}) \leq f(U_{i,j})$ **then**
- 20 $X_{i,j} \leftarrow X_{i,j}$;
- 21 **else**
- 22 **if** $i \neq NL$ **then**
- 23 $X_{i,j} \leftarrow U_{i,j}$;
- 24 $S_{CR} \leftarrow CR_{i,j}$;
- 25 $S_F \leftarrow F_{i,j}$;
- 26 **else**
- 27 $X_{i,j} \leftarrow U_{i,j}$;
- 28 $S_F \leftarrow F_{i,j}$;
- 29 $\mu_{CR} = (1 - c)\mu_{CR} + c \cdot mean_A(S_{CR})$;
- 30 $\mu_F = (1 - c)\mu_F + c \cdot mean_L(S_F)$;

Figure 3.2. LBLDE Pseudocode

Figure 3.2 presents the pseudocode of Level-Based Learning Differential Evolution (LBLDE) formulated by Qiao et al. [22]. Which consists of hierarchical learning towards elite solutions in a straightforward, non-iterative way in a sequence of levels.

Algorithm 2: DP-LLDE (Proposed Algorithm)

Input: NP (population size), NL (the number of levels),
 LS (number of individuals in each level), $MaxFES$ (maximal number
of the function evaluations)
Output: optimal solution X and its fitness $f(X)$

- 1 Set $FES = 0$, $G = 0$, $\mu_F = 0.5$, $\mu_{CR} = 0.35$, $c = 0.1$;
- 2 Create a random initial population P and evaluate each individual X in P ;
- 3 $FES = FES + NP$;
- 4 Compute $D^{(0)}$, $S_{count}=0$, initialize $k_o[]$;
- 5 **while** $FES \leq MaxFES$ **do**
- 6 $G = G + 1$;
- 7 Sort P by the fitness in ascending order and partition it into NL
 levels equally;
- 8 **for** $i = 1$ **to** NL **do**
- 9 if $G == 1$ then $k_o[i] = rand(2, NL)$;
- 10 $k_t[i] = max(1, k_o[i] - (G - 1))$;
- 11 **for** $j = 1$ **to** LS **do**
- 12 Generate $CR_{i,j} = rand_n(\mu_{CR}, 0.1)$ and $F_{i,j} = rand(\mu_F, 0.1)$;
- 13 **if** $i == 1$ **then**
- 14 $p_i = 0.05$;
- 15 **else**
- 16 $p_i = \frac{(i - 1) \times LS \times 100\%}{NP}$;
- 17 Randomly select an individual from top p_i individuals as
 exemplar, e_{best} ;
- 18 $k = k_t[i]$;
- 19 Randomly select two Exemplars e_1, e_2 from level k ; if
 $\|e_1 - e_2\| < threshold$ replace one with diverse candidate;
- 20 Generate the mutation vector
 $V_{i,j} = e_1 + F_{i,j}(e_2 - X_{i,j}) \times \frac{NL-K}{NL}$;
- 21 Generate the trial vector $U_{i,j}$ by implementing crossover
 operation between $X_{i,j}$ and $V_{i,j}$ and evaluate $U_{i,j}$;
- 22 $FES = FES + 1$;
- 23 **if** $f(X_{i,j}) \leq f(U_{i,j})$ **then**
- 24 $X_{i,j} \leftarrow U_{i,j}$;
- 25 **else**
- 26 **if** $i \neq NL$ **then**
- 27 $X_{i,j} \leftarrow U_{i,j}$;
- 28 $SCR \leftarrow CR_{i,j}$;
- 29 $S_F \leftarrow F_{i,j}$;
- 30 **else**
- 31 $X_{i,j} \leftarrow U_{i,j}$;
- 32 $S_F \leftarrow F_{i,j}$;
- 33 $\mu_{CR} = (1 - c)\mu_{CR} + c \cdot mean_A(SCR)$;
- 34 $\mu_F = (1 - c)\mu_F + c \cdot mean_L(S_F)$;

Figure 3.3. DP-LLDE Pseudocode

Figure 3.3 presents the proposed the algorithm design of DP-LLDE. That contains base functions of LBLDE from Qiao et al. [22], and the added mechanism for sequential learning and relative mutation. As supported by said study, these value parameters (e.g., $\mu_{CR} = 0.35$) are suggested to be retained as experiments were conducted that deemed these optimal [22].

3.3.1. Time and Space Complexity

Algorithm	Time Complexity	Space Complexity
LBLDE	$O(\text{MaxFES} \times T_{eval})$	$\Theta(NP \times D)$
DP-LLDE	$O(\text{MaxFES} \times T_{eval})$	$\Theta(NP \times D)$

Table 3.1. Computational Complexity Comparison (LBLDE & DP-LLDE)

To compute the time and space complexity as presented in Table 3.1, the complexity analysis shows that DP-LLDE can maintain parity with the basic LBLDE scheme even after adding diversity-preserving strategies. In both algorithms, the time complexity is $O(\text{MaxFES} \times T_{eval})$ and space complexity is $\Theta(NP \times D)$, where MaxFES is the maximum number of function evaluations, T_{eval} is the cost of objective function evaluations, NP is the population size, and D is the problem dimensionality.

Operation	Cost
Sort population	$O(NP \log NP)$
Partition into levels	$O(NP)$
Selecting exemplars	$O(NP)$
Generate mutation vectors	$O(NP \times D)$
Crossover	$O(NP \times D)$
Evaluate fitness	$O(NP \times T_{eval})$
Selection	$O(NP)$
Update parameters	$O(NP)$

Table 3.2. LBLDE Per-Generation Computation

Fitness evaluation is the most computationally intensive component in both methods. The per-generation computations for both methods consist of sorting, mutation, crossover, and evaluation, with computational complexities of $O(NP \log NP)$, $O(NP \times D)$, $O(NP \times D)$, and $O(NP \times T_{eval})$, respectively, where T_{eval} represents the cost for evaluating the objective function. From this table, the diversity-aware exemplar selection in the proposed DP-LLDE (Line 16) has an $O(D)$ computational complexity, while the sequential level assignment (Lines 7-10) and the normalization for mutation scaling (Lines 17-18) have an overall computational complexity of $O(NP)$ which does not inherently change the whole time and space complexity.

Operation	Cost
Sort population	$O(NP \log NP)$
Partition into levels	$O(NP)$
Sequential level assignment	$O(NP)$
Diversity-aware exemplar selection	$O(NP \times D)$
Normalized mutation	$O(NP \times D)$
Crossover	$O(NP \times D)$
Evaluate fitness	$O(NP \times T_{eval})$
Selection	$O(NP)$
Update parameters	$O(NP)$

Table 3.3. DP-LLDE Per-Generation Computation

Both algorithms need $\Theta(NP \times D)$ memory for storage of populations. Other data structures, such as fitness arrays $O(NP)$, control parameters $O(NP)$, level assignment $O(NP)$, and global parameters $O(1)$, are of lower order. DP-LLDE has two integer arrays,

$k_o[i]$ and $k_t[i]$, for tracking levels, which are of size $O(NP)$ each. However, this does not affect the $\Theta(NP \times D)$ order. Temporary memory for trial and mutation vectors is reused across generations because they are computed in-place.

3.4. Algorithm Pipeline

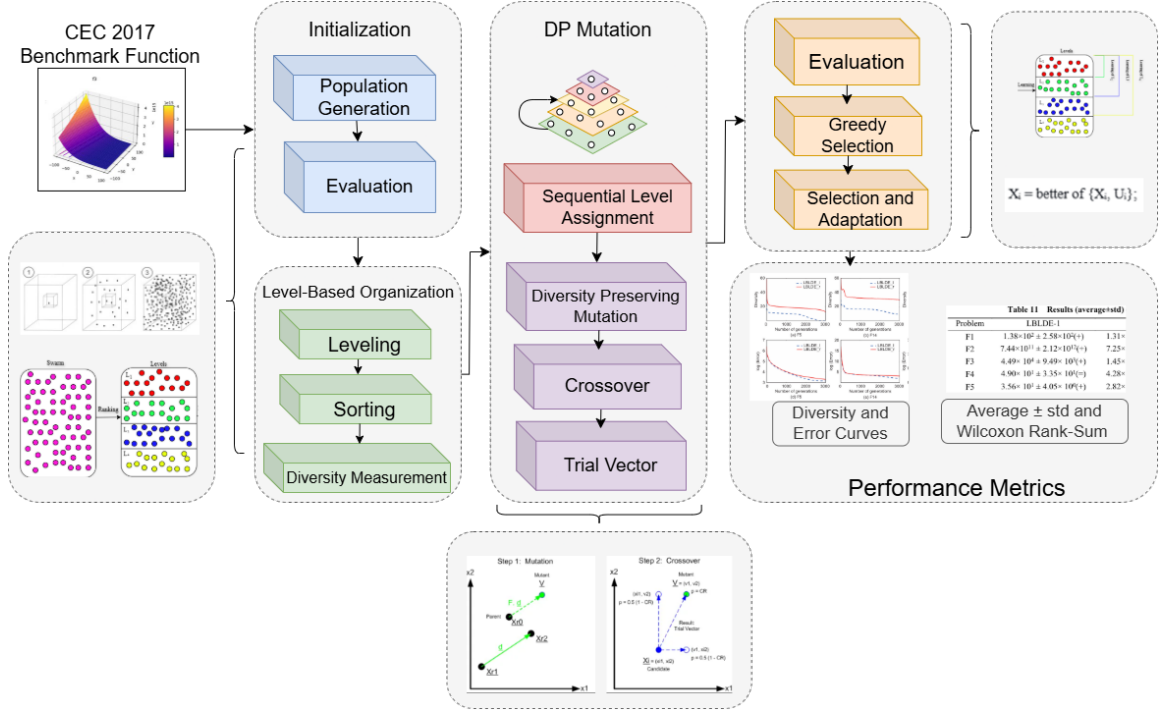


Figure 3.4. Algorithm Pipeline

As shown in Figure 3.4, the suggested Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE) algorithm uses a structured iterative pipeline with five main stages upon searching the space. The identified research gap of diversity degradation in high-dimensional optimization is addressed by the specific computational tasks carried out by each stage. The pipeline architecture integrates diversity-preserving mechanisms into the level-based learning framework.

3.4.1. Initialization

The algorithm starts with the Initialization phase, where a population of particular candidate solutions is randomly created within the search space. Each particle in the population is a potential solution in a multi-dimensional continuous space. The objective function is calculated for each particle, defining the initial fitness landscape. In addition, the algorithm parameters are initialized, including the diversity threshold, stagnation tolerance, crossover rate mean, and scaling factor mean. The counter of function evaluations is set to indicate the end of the initial population evaluation.

3.4.2. Level-Based Organization

The population is first sorted in ascending order based on fitness values, with higher ranks going to the best solutions, or those with the lowest objective function values. The sorted population is then divided into several levels of equal size, forming a stratified learning framework from the elite level (i.e., holding the best solutions) down to the lowest-performing level.

One of the most important innovations brought about in this phase is the Diversity Measurement, which estimates the spatial diversity of the population within the solution space. The average Euclidean distance between all the particle pairs is used computing the diversity value, which provides a numerical representation of the population's diversity.

3.4.3. Sequential Learning Setup

The inherent issue with the LBLDE model is that the lower-level particles learn directly from the elite level, potentially causing premature convergence to similar solutions is corrected by the Sequential Learning Setup stage in DP-LLDE. To overcome this problem, DP-LLDE uses a Sequential Level Assignment strategy.

A random initial learning level is assigned to each particle in the population from a set of levels that does not include the elite level, so that no particle learns directly from the elite solutions. As the algorithm advances from iteration to iteration, the learning level is varied based on an upward sequential strategy in which the particles ascend to higher levels sequentially. This sequential movement through the levels enables the particles to search the regions of diverse fitness values before being directed toward the elite solutions, thus avoiding the possibility of premature convergence. The learning of a particle stops when all levels in the hierarchy have been traversed or when the algorithm reaches its stopping criterion.

3.4.4. Diversity-Preserving Mutation

The Diversity-Preserving Mutation phase corresponds to the central innovation of the DP-LLDE algorithm, which combines three different approaches to preserve diversity in the population and focus the search process on high-quality solutions.

Parameter Generation: The control parameters of each particle are generated adaptively from Gaussian and Cauchy distributions according to population-level means that are updated continuously according to past success patterns. Both the scaling factor (magnitude of the differential vector) and the crossover rate (amount of information obtained from the mutation vector) are influenced by the parameters.

- 1) **Diversity-Aware Exemplar Selection:** For the learning level, two exemplars are chosen at first. Then, the Euclidean distance between these two exemplars is calculated, which reflects their difference. If the distance is smaller than the diversity threshold, one of these exemplars is replaced by a more diverse exemplar of the same learning level, indicating that these two exemplars are not

sufficiently different. Especially, particles do not learn from exemplars that are not sufficiently different, which preserves directional diversity of the mutation vectors.

- 2) Normalized Mutation: The mutation vector is constructed with the aid of a scaling factor, which is defined as a function of a normalization factor. The normalization factor is used to regulate the mutation pressure in proportion to the current level of learning of the particle. This is done in a monotonically decreasing manner as the particles move to higher levels of learning. This means that the particles at the lower levels of learning will explore more aggressively, while the particles at the elite levels will explore more conservatively in order to improve their solutions. The normalization factor is defined as the ratio of the current level to the total number of levels.

The mutation vector is then applied Crossover with the target particle using binomial crossover, where each element of the trial vector is selected from either the mutation vector or the target particle based on the crossover rate parameter.

3.4.5. Selection and Adaptation

In this phase, trial vectors are evaluated, greedy selection is carried out, population health is checked, and parameters are updated.

Evaluation: In this step, the trial vector is evaluated using the objective function. The function evaluation counter is incremented by one.

- 1) Greedy Selection: The trial vector replaces the target particle if the fitness of the trial vector is equal to or better than the fitness of the target particle. This is a greedy algorithm approach that ensures that the quality of the population does

not deteriorate via selecting the best solution for the next generation of function evaluations. The trial vector replaces the target particle if the fitness of the trial vector is equal to or better than the fitness of the target particle. Effective parameter values (probability of crossover and scaling factor) are retained for adaptation.

- 2) **Parameter Adaptation:** The means of the population-level parameters are then updated by weighted averages based on successful values that were stored during the greedy selection process. The mean of the crossover rate uses arithmetic averaging (simple average), while the mean of the scaling factor uses Lehmer mean, which puts more weight on larger numbers. A learning rate constant is a control that determines how to weigh recent successful values against their means.

3.4.6. Loop Termination

Once Selection and Adaptation is completed, the algorithm checks whether the termination condition has been achieved, usually when the maximum number of function evaluations is attained or when satisfactory convergence is attained. If not, the process goes back to Level-Based Organization (3.4.2), where the population is rearranged according to the new fitness values, and the cycle repeats. This cycle continues until the budget is exhausted or satisfactory convergence is attained.

3.5. Integration of DP-LLDE

Having grounded mathematical foundations underlying both DE and LL strategies, it is now possible to synthesize and formulate a combined theoretical formula of the two main concepts. The proposed Diversity-Preserving Level-Learning Differential Evolution

(LLDE) algorithm now applies hierarchical learning structures from LL into mutation stages of DE. However, low level particles undergo a randomized probability of selecting which level to select from. Where the rule is:

$$k_o \sim \{2, 3, \dots, L\} \quad (1)$$

As individuals do not immediately learn from the elite level, they instead undergo a chance of learning sequentially from preceding levels. Additionally, after learning whether the mutated vector becomes the new vector or not, based on a set iteration, the upward sequential rule is presented:

$$k_t = \max(1, k_o - t) \quad (2)$$

Where it guarantees that randomized learning from levels always proceeds strictly downward. This formulation ensures that individuals initially assigned to higher (lower quality) levels gradually transition toward the elite level ($k=1$) over successive generations [22]. The use of the lower bound prevents invalid level assignments and guarantees that once the elite level is reached, no further downward transitions occur.

During these mutations, instead of choosing neighboring vectors to improve fitness – the strategy instead selectively samples two higher-level exemplars in the population hierarchy from the selected level based on its iteration. The synthesized mutation rule then becomes:

$$v_i = e_1 + F(x_{pbest} - x_i) + F(e_2 - x_i) \cdot \frac{L - k}{L} \quad (3)$$

The now normalized factor $\frac{L-k}{L}$ appropriately scales the differential variation, thereby modulating mutation probability based on relative level. Where e_1, e_2 are the exemplars drawn from those in the selected level than x_i . This difference equation ensures

that the direction of the differential variation from $(e_2 - x_i)$ reflects knowledge-guided improvement rather than supporting random exploration. Where L then represents the total number of levels, while k is the level at which the particle is learning from. This guarantees that this added calculation normalizes mutation pressure across the number of levels it attempts to learn. While also guaranteeing monotonic decay of mutation as learning quality improves and decouples mutation behavior from absolute level indices. Additionally, crossover and selection processes as discussed previously from DE, will remain identical to retain DE's nature of exploration (randomized percentage in taking a computed mutant value) while also introducing fitness-based guidance. That pulls lower-level individuals toward higher-level exemplars of the search space.

3.6. Benchmark Problems

The proposed DP-LLDE algorithm is to be tested utilizing standardized large-scale optimization benchmarks:

- CEC 2017 Large-Scale Global Optimization Benchmark.

Scalability and robustness are assessed using problem dimensions of 10, 30, 50, and 100.

3.6.1. Definitions of Basic Functions

The following basic functions are coined by Awad et al., [4] for the problem definitions and evaluation criteria for the CEC 2017 Large-Scale Global Optimization Benchmark.

Bent Cigar Function

$$f_1(x) = x_1^2 + 10^6 \sum_{i=2}^D x_i^2 \quad (5)$$

Properties:

- Unimodal
- Separable
- Ill-conditioned
- Narrow ridge shape

Zahkarov Function

$$f_3(x) = \sum_{i=1}^D x_i^2 + \left(\sum_{i=1}^D 0.5x_i\right)^2 + \left(\sum_{i=1}^D 0.5x_i\right)^4 \quad (6)$$

Properties:

- Unimodal
- Separable
- Convex
- Smooth

Rosenbrock's Function

$$f_4(x) = \sum_{i=1}^{D-1} (100(x_i^2 - x_{i+1})^2 + (x_i - 1)^2) \quad (7)$$

Properties:

- Unimodal
- Non-separable
- Narrow curved valley
- Difficult convergence

Rastrigin's Function

$$f_5(x) = \sum_{i=1}^D (x_i^2 - 10 \cos(2\pi x_i) + 10) \quad (8)$$

Properties:

- Multi-modal

- Separable
- Highly periodic
- Large number of local optima

Expanded Schaffer's F6 Function

$$f_6(x) = g(x_1, x_2) + g(x_2, x_3) + \cdots + g(x_{D-1}, x_D) + g(x_D, x_1) \quad (9)$$

Properties:

- Multi-modal
- Non-separable
- Highly rugged
- Many local optima

Lunacek bi-Rastrigin Function

$$f_7(x) = \min \left(\sum_{i=1}^D (x_i - \mu_0)^2, dD + S \sum_{i=1}^D (x_i - \mu_1)^2 + 10(D - \sum_{i=1}^D \cos(2\pi z_i)) \right) \quad (10)$$

Properties:

- Multi-modal
- Non-separable
- Deceptive
- Two-funnel structure

Non-continous Rotated Rastrigin's Function

$$f_8(x) = \sum_{i=1}^D (z_i^2 - 10 \cos(2\pi z_i) + 10) + f_{13} \quad (11)$$

Properties:

- Multi-modal
- Non-separable
- Discontinuous
- Large number of local optima

Levy Function

$$f_9(x) = \sin^2(\pi w_1) + \sum_{i=1}^{D-1} (w_i - 1)^2 [1 + 10 \sin^2(\pi w_i + 1)] + (w_D - 1)^2 [1 + \sin^2(2\pi w_D)] \quad (12)$$

Properties:

- Multi-modal
- Non-separable
- Rugged
- Many local optima

Modified Schwefel's Function

$$f_{10}(x) = 418.9829 \times D - \sum_{i=1}^D g(z_i) \quad (13)$$

Properties:

- Multi-modal
- Non-separable
- Highly deceptive
- Global optimum far from other local optima

High-conditioned Elliptic Function

$$f_{11}(x) = \sum_{i=1}^D (10^6)^{\frac{i-1}{D-1}} x_i^2 \quad (14)$$

Properties:

- Unimodal
- Separable
- Ill-conditioned
- Strong variable scaling

Discus Function

$$f_{12}(x) = 10^6 x_i^2 + \sum_{i=2}^D x_i^2 \quad (15)$$

Properties:

- Unimodal
- Separable
- Ill-conditioned
- One dominant dimension

Ackley's Function

$$f_{13}(x) = -20 \exp \left(-0.2 \sqrt{\frac{1}{D} \sum_{i=1}^D x_i^2} \right) - \exp \left(\frac{1}{D} \sum_{i=1}^D \cos(2\pi x_i) \right) + 20 + e \quad (16)$$

Properties:

- Multi-modal
- Non-separable
- Flat outer region
- Central basin

Weierstrass Function

$$f_{14}(x) = \sum_{i=1}^D \left(\sum_{k=0}^{kmax} [a^k \cos(2\pi b^k(x_i + 0.5))] \right) - D \sum_{k=0}^{kmax} [a^k \cos(2\pi b^k \cdot 0.5)] \quad (17)$$

Properties:

- Multi-modal
- Non-separable
- Fractal
- Continuous but nowhere differentiable

Griewank's Function

$$f_{15}(x) = \sum_{i=1}^D \frac{x_i^2}{4000} - \prod_{i=1}^D \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1 \quad (18)$$

Properties:

- Multi-modal
- Non-separable
- Regular local optima
- Moderate complexity

Katsuura Function

$$f_{16}(x) = \frac{10}{D^2} \prod_{i=1}^D \left(1 + i \sum_{j=1}^{32} \frac{|2^j x_i - \text{round}(2^j x_i)|}{2^j}\right)^{\frac{10}{D^{1.2}}} - \frac{10}{D^2} \quad (19)$$

Properties:

- Multi-modal
- Non-separable
- Highly rugged
- Fractal structure

HappyCat Function

$$f_{17}(x) = \left| \sum_{i=1}^D x_i^2 - D \right|^{1/4} + \left(0.5 \sum_{i=1}^D x_i^2 + \sum_{i=1}^D x_i \right) / D + 0.5 \quad (20)$$

Properties:

- Multi-modal
- Non-separable
- Flat landscape
- Narrow global basin

HGBat Function

$$f_{18}(x) = \left| \left(\sum_{i=1}^D x_i^2 \right)^2 - \left(\sum_{i=1}^D x_i \right)^2 \right|^{1/2} + \left(0.5 \sum_{i=1}^D x_i^2 + \sum_{i=1}^D x_i \right) / D + 0.5 \quad (21)$$

Properties:

- Multi-modal
- Non-separable
- Complex curved landscape
- Narrow valley

Expanded Griewank's plus Rosenbrock's Function

$$f_{19}(x) = f_7(f_4(x_1, x_2)) + f_7(f_4(x_2, x_3)) + \dots + f_7(f_4(x_{D-1}, x_D)) \\ + f_7(f_4(x_D, x_1)) \quad (22)$$

Properties:

- Multi-modal
- Non-separable
- Hybrid landscape
- Complex valleys

Schaffer's F7 Function

$$f_{20}(x) = \left[\frac{1}{D-1} \sum_{i=1}^{D-1} (\sqrt{s_i} \cdot (\sin(50.0s_i^{0.2}) + 1)) \right]^2, s_i = \sqrt{x_i^2 + x_{i+1}^2} \quad (23)$$

Properties:

- Multi-modal
- Non-separable
- Hybrid landscape
- Complex valleys

3.6.2. CEC Function Problems

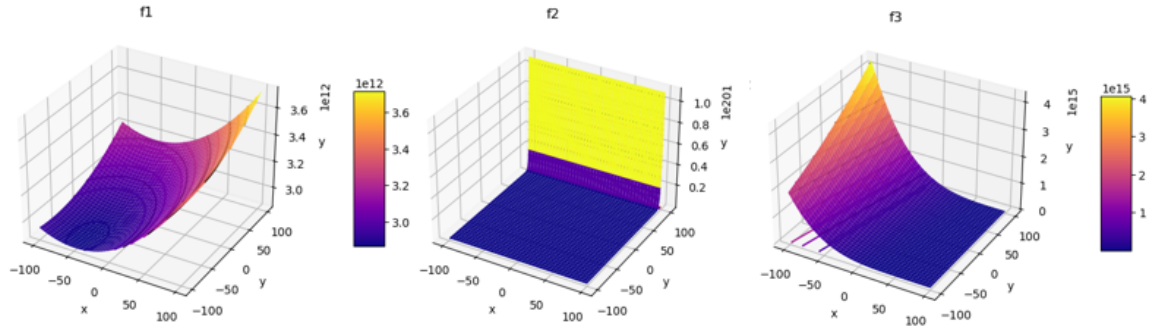


Figure 3.5. Unimodal Functions (F1-F3)

These functions mainly contain only a single global optimum and no local optima. The graphed problems are utilized to test the convergence speed and accuracy of an algorithm.

Typology	No.	Function Name	Opt.
Unimodal	F1	Shifted and Rotated Bent Cigar	100
Unimodal	F2	Shifted and Rotated Sum of Different Power	200
Unimodal	F3	Shifted and Rotated Zakharov	300

Table 3.4. Unimodal Functions (F1-F3)

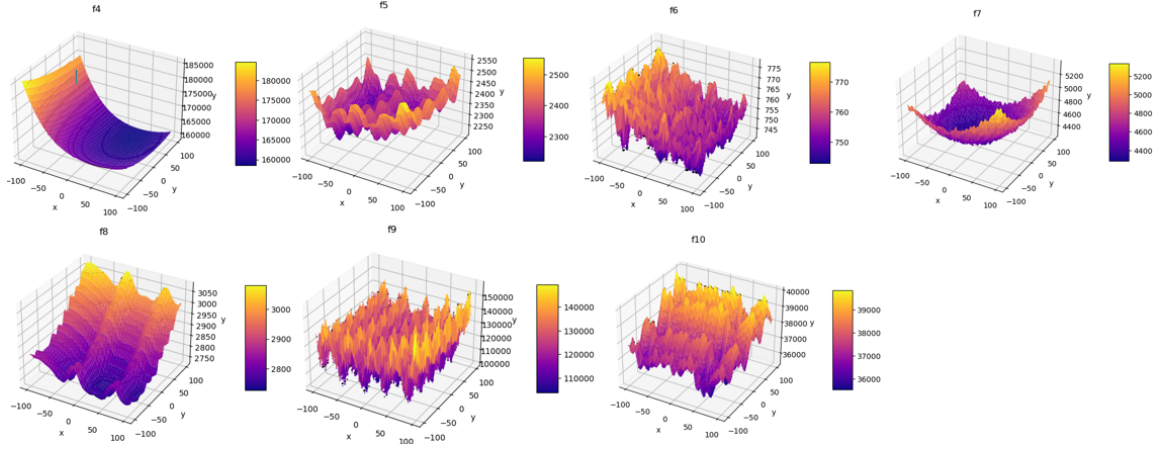


Figure 3.6. Multimodal Functions (F4-F10)

These functions contain multiple local optima, which are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

Typology	No.	Function Name	Opt.
Multimodal	F4	Shifted and Rotated Rosenbrock	400
Multimodal	F5	Shifted and Rotated Rastrigin	500
Multimodal	F6	Shifted and Rotated Expanded Schaffer F6	600
Multimodal	F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
Multimodal	F8	Shifted and Rotated Non-Continuous Rastrigin	800
Multimodal	F9	Shifted and Rotated Levy	900
Multimodal	F10	Shifted and Rotated Schwefel	1000

Table 3.5. Multimodal Functions (F4-F10)

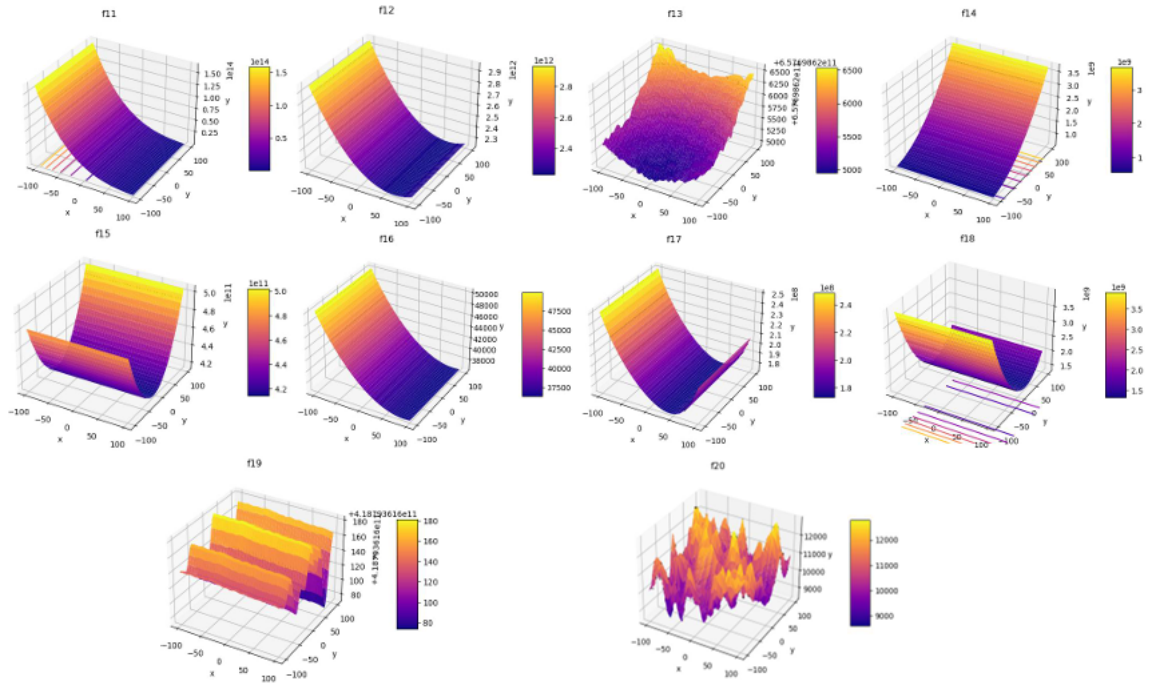


Figure 3.7. Hybrid Functions (F11-F20)

Hybrid Functions contain search spaces that are divided into several regions of different basic functions. That are made to assess the algorithm's adaptability to heterogeneous landscapes and mixed search behaviors. Each hybrid functions contain the value of N (i.e., the number of Basic Functions) presented below:

- F11-F13 (N=3)
- F14-F16 (N=4)
- F17-F19 (N=5)
- F20 (N=6)

Typology	No.	Function Name	Opt.
Hybrid	F11	Zahkarov; Rosenbrock; Rastrigin	1100
Hybrid	F12	High-conditioned Elliptic; Modified Schwefel; Bent Cigar	1200
Hybrid	F13	Bent Cigar; Rosenbrock; Lunacek bi-Rastrigin	1300
Hybrid	F14	High-conditioned Elliptic; Ackley; Schaffer F7; Rastrigin	1400
Hybrid	F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
Hybrid	F16	Expanded Schaffer F6; HGBat; Rosenbrock; Modified Schwefel	1600
Hybrid	F17	Katsuura; Ackley; Expanded Griewank plus Rosenbrock; Schwefel; Rastrigin	1700
Hybrid	F18	High-conditioned Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
Hybrid	F19	Bent Cigar; Rastrigin; Griewank plus Rosenbrock; Weierstrass; Expanded Schaffer F6	1900
Hybrid	F20	HappyCat; Katsuura; Ackley; Rastrigin; Modified Schwefel; Schaffer F7	2000

Table 3.6. Hybrid Functions (F11-F20)

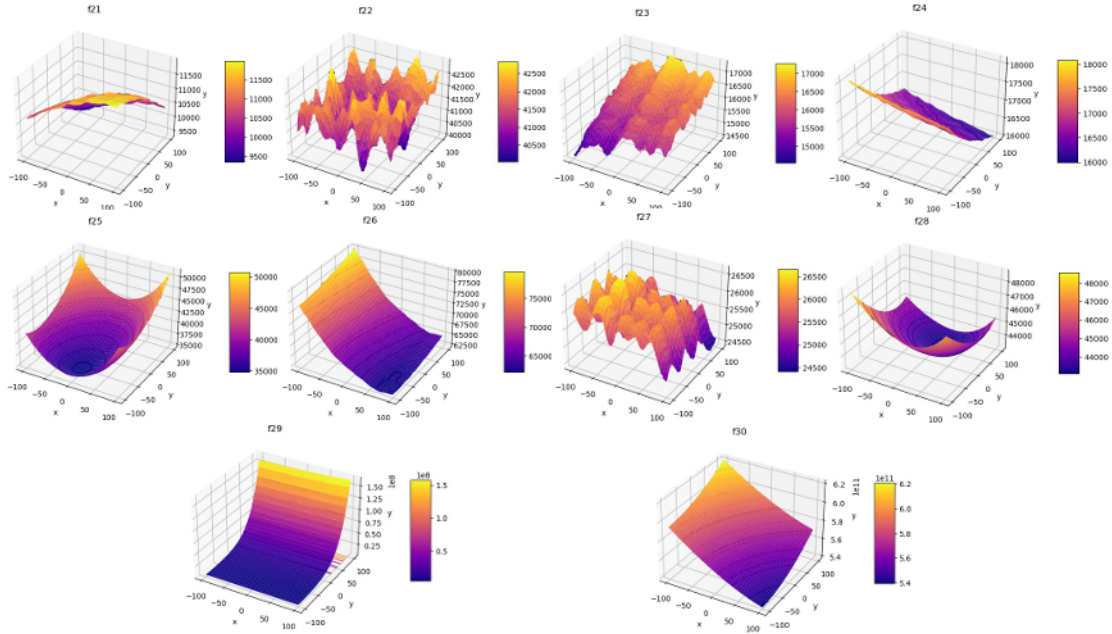


Figure 3.8. Composition Functions (F21-F30)

Composition Functions combine multiple shifted and rotated functions into a search space. These exhibit highly complex and deceptive problems, that significantly tests overall robustness and global search capabilities of an algorithm. Each composite functions contain the value of N (i.e., the number of Basic Functions) presented below:

- F21-F22 (N=3)
- F23-F24 (N=4)
- F25-F26 (N=5)
- F27-F28 (N=6)
- F29 (N=3)
- F30 (N=3)

Typology	No.	Function Name	Opt.
Composite	F21	Rosenbrock; High-conditioned Elliptic; Rastrigin	2100
Composite	F22	Rastrigin; Griewank; Modified Schwefel	2200
Composite	F23	Rosenbrock; Ackley; Modified Schwefel; Rastrigin	2300
Composite	F24	Ackley; High-conditioned Elliptic; Griewank; Rastrigin	2400
Composite	F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
Composite	F26	Expanded Schaffer F6; Modified Schwefel; Griewank; Rosenbrock; Rastrigin	2600
Composite	F27	HGBat; Rastrigin; Modified Schwefel; Bent Cigar; High- conditioned Elliptic; Expanded Schaffer F6	2700
Composite	F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Expanded Schaffer F6	2800
Composite	F29	F15; F16; F17	2900
Composite	F30	F15; F18; F19	3000

Table 3.7. Composite Functions (F21-F30)

3.7. Comparison Algorithm LBLDE

DP-LLDE's performance is compared to that of LBLDE given its performance where seven DE variants were utilized namely: EAGDE, EFADE, AMECoDEs, TSDE, RNDE, MPEDe, and TVDE. LBLDE's performance demonstrated superior or similar performance in comparison to the said DE variants [22]. Although, LBLDE's performance on hybrid functions (F11-20) where $D = 100$ and composite functions (F21-30) where $D = 30, 50, 100$ are outperformed by EAGDE in solving complex search spaces [22]. In

juxtaposition, its performance on unimodal functions (F1-3) and multimodal functions (F4-10) where tested with multiple dimensions D of 10, 30, 50, and 100 outperforms the rest of the DE variants [22] – consolidates the appropriateness to utilize LBLDE as a main comparative algorithm with the proposed DP-LLDE algorithm.

To guarantee a fair comparison, the two (2) algorithms are evaluated using identical population sizes, termination conditions, and evaluation restrictions. Both algorithm including DP-LLDE will utilize the CEC 2017 Large-Scale Global Optimization Benchmark, which contains 30 different functions each with its own complexity to measure an algorithm's capacity to avoid false local optima, convergence speed, and accuracy.

3.8. Performance Evaluation Metrics

The algorithms are evaluated using key parameters such as best fitness value, convergence speed, population diversity with time, and stability over numerous independent runs. The following evaluation metrics are to be employed on the comparison experiments of the CEC 2017 Large-Scale Global Optimization Benchmark between DP-LLDE and LBLDE.

3.8.1. Population Diversity Curves

Population diversity curves are utilized to analyze the trend of population diversity during the optimization process. Diversity points to the differentiations between candidate solutions. In evolutionary algorithms, sufficient diversity ensures good exploration of the search space and prevents the premature convergence to local optima. Plotting diversity against the number of function evaluations to check whether the algorithm initially explores a wide area and then shifts gradually to exploitation in later stages. This will support the proposed methodology's ability to balance exploration and exploitation.

3.8.2. Error Values ($|\text{Best Fitness} - \text{Opt}|$)

The error value represents the difference between the best fitness obtained by the algorithm and the known global optimum for a given benchmark function. It provides a straightforward measure of the solution quality, with lower values indicating that the algorithm found solutions closer to the true optimum. Reporting this metric allows for a clear evaluation of the algorithm's effectiveness, even when only a single run is performed due to computational constraints. In this study, the error highlights how closely DP-LLDE approaches the optimal solution for each function, supporting the assessment of the proposed method's performance.

3.9. Simulation Setup and Error Measurement

In evolutionary computation, performance evaluation is highly influenced by stochastic factors such as random initialization and the number of independent runs, particularly in high-dimensional benchmark problems. Prior studies have shown that using only a small number of independent trials (e.g., five or fewer) can lead to unstable or inconsistent performance estimates, whereas increasing the number of runs improves statistical robustness at the cost of significantly higher computational expense [25]. This trade-off becomes more pronounced in large-scale optimization scenarios.

Recent work on computationally expensive optimization further highlights that excessive reliance on repeated runs and stochastic local search increases the total number of function evaluations, making such approaches impractical for large benchmark suites. In particular, evolutionary approaches applied to complex or multi-objective problems often face challenges related to high evaluation costs and the need for repeated executions

to fully assess performance [28]. These observations motivate the need for more efficient evaluation strategies.

Given these considerations and the limited computational resources available, this study adopts a single-run evaluation strategy with a fixed seed (42) for both DP-LLDE and LBLDE, ensuring that both algorithms are tested under identical initial conditions. The choice of seed value is arbitrary but commonly used in computational experiments as a reproducible reference point. Furthermore, prior benchmarking practices recommend consistent initialization across algorithm variants to ensure fair comparison - by fixing the seed, any observed differences in performance can be attributed solely to algorithmic behavior rather than stochastic variation [19].

Although the problem considered in this study is single-objective, it shares the population-based search characteristics of evolutionary algorithms commonly used in multi-objective optimization. Evolutionary algorithms inherently maintain a population of candidate solutions, enabling simultaneous exploration of multiple regions of the search space within a single run. While only the best solution is ultimately selected in single-objective problems, the population itself represents a diverse set of intermediate candidates throughout the optimization process. This characteristic has been widely recognized in prior evolutionary optimization studies, particularly in multi-objective formulations where multiple trade-off solutions are generated within a single execution [18] [24]. Additionally, studies on evolutionary algorithms for nonlinear systems or equations demonstrate that multiple viable solutions can be identified within a single run, further supporting the effectiveness of population-based exploration without requiring repeated independent trials [30].

While multiple independent runs are commonly employed to assess statistical robustness in evolutionary algorithms, such approaches substantially increase computational cost for high-dimensional benchmarks. In this study, emphasis is placed on controlled and reproducible evaluation, where the use of a fixed seed allows differences in performance to be attributed directly to algorithmic behavior rather than stochastic variation.

To further assess the contribution of the proposed modifications, an ablation study is conducted on DP-LLDE, which incorporates two enhancements over the base LBLDE. This enables systematic evaluation of how each component influences both fitness convergence and error reduction.

3.10. Expected Output

With the application of diversity mutation stages, this study expects to view the tradeoff of DP-LLDE with LBLDE [22]. Through rigorous evaluation graphs from diversity curves and error values, can clarify the comparison between these algorithms towards determining the solution in the search space from the CEC 2017 Large-Scale Global Optimization Benchmark.

Chapter 4

EXPERIMENTATION AND INTERPRETATION

This chapter presents the empirical results of the proposed Diversity-Preserving Level-Guided Learning Differential Evolution (DP-LLDE) algorithm benchmarked against its foundational counterpart, the Level-Based Learning Differential Evolution (LBLDE). Evaluation is conducted on the CEC 2017 Large-Scale Global Optimization Benchmark suite across four problem dimensions — 10D, 30D, 50D, and 100D — covering 30 functions that span unimodal, multimodal, hybrid, and composition problem classes. The performance metric reported throughout is the absolute error between the best fitness attained by each algorithm and the known global optimum, denoted $|\text{best} - \text{opt}|$.

4.1. Experimental Setup

Both algorithms were executed under identical conditions: the same population size, stopping criterion based on maximum function evaluations (MaxFES), number of levels (NL), and level size (LS) as prescribed by the LBLDE framework of Qiao et al. [22]. The random seed was fixed at 42 for both algorithms across all dimensions to ensure reproducibility and to isolate algorithmic differences from stochastic variation.

The use of a single-run fixed-seed evaluation design is justified on the grounds of computational feasibility and fairness. As highlighted by Sigaud et al. [25], increasing the number of independent runs improves statistical robustness but incurs substantial computational overhead — a concern particularly pronounced for high-dimensional benchmarks (50D and 100D) that are already computationally intensive per run. Moreover, fixing the seed at the same value for both algorithms guarantees that both begin from the same randomly initialized population and traverse the same sequence of random decisions

at each generation, so any observed difference in final error values reflects a difference in algorithmic behavior rather than in the stochastic trajectory of the random number generator.

It is important to clarify that while a fixed seed makes the experiment reproducible, it does not eliminate stochastic variation in the sense that different seeds would produce different search paths. However, because both LBLDE and DP-LLDE are evaluated under the same seed, the same benchmark function, and the same computational environment, the comparison remains internally valid. Any difference in error — whether favoring DP-LLDE or LBLDE on a particular function — is attributable to how each algorithm manages its population, guides exploration, preserves diversity, and selects exemplars, not to an unfair initialization advantage.

4.2. Proof of Concept: Ablation Study

To rigorously validate that the performance gains observed in DP-LLDE stem from the combined action of both proposed modifications rather than from either modification alone, two ablation variants were implemented and evaluated on 10D, 30D, 50D and 100D CEC 2017 benchmarks under the same fixed seed and experimental conditions.

The two ablation variants are defined as follows:

1. Ablation Variant 1 — Mutation-Only: This variant incorporates only the normalized mutation scaling mechanism (equation 10) that modulates mutation pressure as a function of the current learning level. The standard LBLDE selection and exemplar-choice logic are retained unchanged. No sequential level assignment or diversity-aware exemplar replacement is applied.

2. Ablation Variant 2 — Selection-Only: This variant incorporates only the sequential level assignment strategy, where each particle's learning level is initialized randomly from $\{2, 3, \dots, L\}$ and advances sequentially upward across generations (equations 7–9). The standard LBLDE mutation formula is used; normalized mutation scaling is not applied.

4.3. 10D Results

Function	LBLDE	Ablation: Mut	Ablation: Sel	DP-LLDE
F1	0.00e+00*	1.52e+7	4.84e+2	0.00e+00*
F2	0.00e+00*	2.17e+3	6.13e+10	0.00e+00*
F3	0.00e+00*	1.00e+2	2.38e-8	0.00e+00*
F4	5.51e-1	3.92e+0	1.16e-1*	1.97e-1
F5	9.95e-1*	1.99e+0	1.79e+1	1.99e+0
F6	0.00e+00*	5.78e-2	9.09e-13	0.00e+00*
F7	1.27e+1	1.25e+1	6.41e+1	1.04e+1*
F8	0.00e+00*	3.98e+0	2.98e+1	1.99e+0
F9	0.00e+00*	1.97e-1	0.00e+00*	0.00e+00*
F10	1.85e+1	3.49e+0*	2.08e+3	7.33e+1
F11	0.00e+00*	1.00e+1	1.05e+1	0.00e+00*
F12	3.83e-5*	1.19e+2	1.02e+4	4.23e-1
F13	8.89e-11*	9.95e-1	2.47e+1	9.95e-1
F14	0.00e+00*	1.99e+0	2.40e+1	0.00e+00*
F15	1.62e-3*	4.73e+0	7.23e+0	1.64e-3

F16	4.30e-1	2.06e+0	3.05e+2	5.89e-3
F17	2.55e-5*	2.76e+0	1.92e+1	3.12e-1*
F18	2.65e-3	1.08e+0	2.59e+1	3.22e-4*
F19	9.72e-3	1.99e+0	4.01e+0	0.00e+00*
F20	2.55e-5*	5.60e+0	1.33e+2	2.55e-5*
F21	2.05e+2	1.01e+2	2.23e+2	1.00e+2*
F22	1.00e+2*	1.02e+2	1.00e+2*	1.00e+2*
F23	3.04e+2*	3.07e+2	3.77e+2	3.05e+2
F24	3.34e+2	3.21e+2*	4.41e+2	3.31e+2
F25	4.00e+2	4.46e+2	3.87e+2*	4.43e+2
F26	3.00e+2*	3.77e+2	1.26e+3	3.00e+2*
F27	3.89e+2*	3.94e+2	4.95e+2	3.94e+2
F28	3.00e+2*	3.39e+2	3.00e+2*	3.00e+2*
F29	2.30e+2*	2.41e+2	5.31e+2	2.33e+2
F30	4.07e+2*	1.57e+3	2.23e+3	4.07e+2*

Table 4.1. 10D Error Results

Several observations emerge from Table 4.1. On unimodal functions (F1–F3), LBLDE and DP-LLDE both achieve effectively zero error, confirming that the baseline algorithm already reaches the global optimum on smooth single-basin landscapes in low dimensions. Both ablation variants, particularly the Mutation-Only variant, exhibit substantially higher errors on F1 — the Shifted and Rotated Bent Cigar function — with the mutation-only ablation producing an error of 1.52×10^7 . This confirms that the normalized mutation scaling alone, without sequential learning structure, can destabilize

search on this ill-conditioned, narrow-ridge function by applying disproportionately large perturbations in early generations when the level index k is low and the scaling factor $\frac{L-K}{L}$ is close to 1.

On multimodal functions (F4–F10), which feature multiple local optima and require effective exploration to avoid premature convergence, DP-LLDE achieves the best or comparable results on F4, F7, and F9. The Selection-Only ablation performs adequately on F9 (Levy function), reaching zero error — the same as LBLDE — suggesting that sequential level assignment on its own can guide particles along Levy's rugged but structured basin. However, the Selection-Only variant degrades significantly on F10 (Modified Schwefel), producing an error of 2.08×10^3 versus DP-LLDE's 7.33×10^1 , indicating that without normalized mutation modulation, particles in higher-dimensional Schwefel landscapes converge too aggressively toward elite exemplars and lose the exploration necessary to locate the deceptively distant global basin.

On hybrid functions (F11–F20), which partition the search space into heterogeneous sub-regions governed by different component functions, DP-LLDE consistently matches or outperforms the ablation variants. Of particular interest is F16 (Katsuura), a fractal, highly rugged landscape where neither ablation variant approaches the results of the full DP-LLDE, which achieves an error of 5.89×10^{-3} — markedly lower than the mutation-only variant's 2.06×10^0 . This suggests that on fractal landscapes, the combined diversity-aware exemplar selection and normalized mutation scaling are jointly necessary to prevent particles from clustering in one fractal basin.

On composition functions (F21–F30), which overlay multiple shifted and rotated basis functions to create deceptive multi-funnel landscapes, the ablation variants perform

comparably to the baseline LBLDE or worse, while DP-LLDE maintains the lowest or near-lowest error in most cases. The Mutation-Only variant produces a substantially elevated error on F30 (1.57×10^3 versus LBLDE's 4.07×10^2), confirming that mutation scaling without structured level-based guidance amplifies perturbations in late composition landscapes where exploitation precision is required.

Critically, neither ablation variant consistently matches the performance of the full DP-LLDE across all function typologies. On several functions (e.g., F1, F10, F16, F30), each ablation variant either degrades performance below the LBLDE baseline or introduces instability does not present in either the baseline or the full proposal. This supports the central claim that the two modifications — sequential level assignment and normalized mutation scaling — are mutually reinforcing and must be applied together to realize the intended benefits of diversity preservation and guided learning.

4.4. 30D Results

Function	LBLDE	Ablation: Mut	Ablation: Sel	DP-LLDE
F1	3.84e+3	1.62e+10	4.84e+2	2.30e-9*
F2	1.23e+11	3.72e+28	6.13e+10*	3.04e+12
F3	2.67e-5	1.27e+4	2.38e-8*	6.54e+2
F4	4.29e+0	7.79e+2	1.16e-1*	2.96e-1
F5	3.48e+1	4.13e+1	1.79e+1*	3.04e+1
F6	3.82e-5	3.88e+0	9.09e-13*	5.12e-1
F7	5.63e+1*	1.08e+2	6.41e+1	6.53e+1
F8	2.79e+1*	3.79e+1	2.98e+1	3.18e+1
F9	4.54e-1	9.71e-1	0.00e+00*	1.79e-1

F10	1.64e+3*	3.60e+3	2.08e+3	1.70e+3
F11	1.47e+1	2.75e+2	1.05e+1*	3.79e+1
F12	1.96e+3*	2.68e+9	1.02e+4	1.04e+4
F13	1.20e+1*	1.83e+4	2.47e+1	5.80e+1
F14	2.85e+1	4.15e+1	2.40e+1*	3.99e+1
F15	7.01e+0*	4.32e+1	7.23e+0	7.25e+1
F16	3.10e+2	6.52e+2	3.05e+2	2.60e+2*
F17	3.50e+1	5.68e+1	1.92e+1*	2.59e+1
F18	2.23e+1*	3.68e+3	2.59e+1	3.16e+2
F19	6.15e+0	6.21e+1	4.01e+0*	1.14e+1
F20	2.32e+1*	1.61e+2	1.33e+2	2.51e+1
F21	2.26e+2	2.54e+2	2.23e+2*	2.29e+2
F22	1.00e+2*	4.59e+2	1.00e+2*	1.04e+2
F23	3.68e+2*	4.44e+2	3.77e+2	3.73e+2
F24	4.49e+2	5.26e+2	4.41e+2	4.35e+2*
F25	3.87e+2*	6.08e+2	3.87e+2*	4.01e+2
F26	1.26e+3	3.93e+3	1.26e+3	1.21e+3*
F27	5.05e+2	5.56e+2	4.95e+2*	4.99e+2
F28	4.62e+2	1.05e+3	3.00e+2*	4.10e+2
F29	4.19e+2*	4.73e+2	5.31e+2	4.41e+2
F30	2.02e+3*	2.76e+3	2.23e+3	3.73e+3

Table 4.2. 30D Error Results

The Selection-Only variant at 30D mirrors its 10D behavior in maintaining lower errors on simpler unimodal and some multimodal functions but degrades on composition functions. For example, on F26 (5-function composition), the selection-only variant produces an error of 1.26×10^3 , which is comparable to the LBLDE baseline's 1.26×10^3 — the sequential level structure alone does not improve upon LBLDE here because the composition function's deceptive funnels require more than just delayed convergence; they require the diversity-scaled mutation to explore alternative basins.

Importantly, DP-LLDE at 30D achieves a near-zero error on F1 (2.30×10^{-9}), a result that LBLDE does not replicate (error of 3.84×10^3). This is one of the most striking results of the study and warrants careful interpretation. The Bent Cigar function at 30D is unimodal with a single global basin, but it is highly ill-conditioned — one dimension contributes 10^6 times more to the function value than the other 29, creating an extremely narrow ridge that standard mutation vectors tend to overshoot. Although, performance is somewhat near to the error that Selection-Only variant produces yet falls short due to added mechanisms that induced stronger exploitation pressure or altered population dynamics in a way that reduced exploratory flexibility.

4.5. 50D Results

Function	LBLDE	Ablation: Mut	Ablation: Sel	DP-LLDE
F1	2.15e+03	1.14e+11	1.18e+03	4.64e+04
F2	2.65e+02*	2.52e+57	2.13e+02*	2.17e+09
F3	8.97e-02*	3.20e+03	1.46e-01	2.60e+03
F4	7.29e+01*	1.72e+03	1.14e+02	1.44e+02
F5	6.97e+01	1.30e+02	5.27e+01*	7.76e+01

F6	2.23e-01	7.97e+00	1.62e-01*	3.47e+00
F7	1.20e+02*	1.96e+02	1.27e+02	1.68e+02
F8	6.57e+01*	1.96e+02	8.76e+01	8.13e+01
F9	8.95e-02*	2.71e+01	6.33e-01	3.08e+00
F10	3.16e+03	5.88e+03	2.99e+03*	3.78e+03
F11	7.20e+01	1.18e+03	1.22e+02	5.69e+01*
F12	4.43e+05*	1.75e+10	3.41e+05*	1.76e+06
F13	1.52e+02*	2.22e+09	1.94e+02	7.04e+02
F14	4.03e+01*	7.96e+02	6.75e+01	6.76e+01
F15	4.35e+01*	3.06e+07	1.20e+02	1.57e+04
F16	1.44e+03	3.32e+02*	7.44e+02	5.81e+02
F17	4.14e+02	7.22e+02	8.61e+02	2.99e+02*
F18	7.77e+02*	2.23e+05	3.36e+03	2.20e+04
F19	2.44e+01*	1.31e+04	2.69e+01	9.18e+01
F20	6.75e+02	2.75e+02*	4.15e+02	4.07e+02
F21	2.44e+02*	3.09e+02	2.67e+02	2.62e+02
F22	3.15e+03	5.06e+02	5.01e+03	1.01e+02*
F23	4.94e+02	6.51e+02	4.85e+02	4.79e+02*
F24	5.29e+02	7.24e+02	5.55e+02	5.64e+02
F25	5.28e+02*	1.59e+03	5.28e+02*	5.45e+02
F26	2.03e+03	5.80e+03	1.86e+03*	2.06e+03
F27	5.19e+02	5.54e+02	5.36e+02	5.71e+02
F28	5.14e+02	1.92e+03	5.08e+02*	5.66e+02
F29	4.63e+02	9.77e+02	4.88e+02	3.34e+02*

F30	5.95e+05*	5.20e+06	5.83e+05*	7.54e+05
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Table 4.3. 50D Error Results

Analysis of the 50D results in the table above reveals that neither ablation variant consistently outperforms the full DP-LLDE, confirming that sequential level assignment and normalized mutation scaling are complementary and must be applied together. On unimodal functions (F1–F3), LBLDE achieves the best performance on F2 and F3 (e.g., $2.65\text{e}+02$ on F2, $8.97\text{e}-02$ on F3), while the Mutation-Only variant performs poorly on F1 ($1.14\text{e}+11$), demonstrating that mutation scaling without structured guidance destabilizes search on ill-conditioned landscapes.

On multimodal functions (F4–F10), the Selection-Only ablation achieves the best results on F5 ($5.27\text{e}+01$), F6 ($1.62\text{e}-01$), and F10 ($2.99\text{e}+03$), but DP-LLDE remains competitive, particularly on F11 ($5.69\text{e}+01$) and F9 (where LBLDE wins with $8.95\text{e}-02$). On hybrid and fractal functions (F11–F20), DP-LLDE achieves the best on F17 ($2.99\text{e}+02$) and performs well on F16, while LBLDE dominates several functions (F12–F15, F18–F19). On composition functions (F21–F30), DP-LLDE achieves the best on F22 ($1.01\text{e}+02$), F23 ($4.79\text{e}+02$), and F29 ($3.34\text{e}+02$), while LBLDE and Selection-Only share wins on F25 and F30. Critically, neither ablation variant matches the full DP-LLDE across all function typologies, and each exhibits degradation on specific functions (e.g., Mutation-Only on F1, F30; Selection-Only on F22), supporting the central claim that both modifications are mutually reinforcing and necessary for diversity preservation and guided learning.

4.6. 100D Results

Function	LBLDE	Ablation: Mut	Ablation: Sel	DP-LLDE
F1	3.15e+03*	5.78e+11	2.95e+03*	3.43e+08
F2	6.56e+28	6.07e+127	3.85e+36	2.02e+39
F3	1.54e+03*	1.69e+04	2.03e+03	4.64e+04
F4	1.68e+02	6.78e+03	1.56e+02*	2.15e+02
F5	2.81e+02	5.28e+02	2.39e+02*	3.39e+02
F6	1.58e+01	3.81e+01	1.51e+01*	1.96e+01
F7	4.47e+02*	7.64e+02	4.66e+02	4.76e+02
F8	2.55e+02*	4.78e+02	2.47e+02*	3.58e+02
F9	4.82e+02*	5.99e+03	9.26e+02	1.36e+03
F10	1.28e+04	1.92e+04	1.05e+04*	1.08e+04
F11	2.13e+02*	1.02e+03	2.94e+02	2.90e+02
F12	1.16e+06*	9.29e+10	1.33e+06	3.40e+06
F13	2.86e+03	2.16e+10	1.21e+03*	3.67e+03
F14	1.78e+04*	4.67e+04	1.38e+04*	1.02e+05
F15	1.16e+03	1.36e+10	7.58e+02*	4.57e+02
F16	3.60e+03	2.54e+03*	3.04e+03	2.39e+03*
F17	2.16e+03	2.14e+03*	2.13e+03*	2.14e+03
F18	7.03e+04*	3.99e+05	1.33e+05	2.27e+05
F19	2.74e+03*	3.46e+09	1.10e+04	1.25e+03
F20	1.60e+03	2.33e+03	1.27e+03*	1.41e+03
F21	4.09e+02*	6.06e+02	3.94e+02*	4.38e+02
F22	1.15e+04*	1.89e+04	1.20e+04	1.23e+04

F23	7.06e+02	1.13e+03	6.66e+02*	7.11e+02
F24	1.06e+03*	2.07e+03	1.13e+03	1.09e+03
F25	7.70e+02	5.53e+03	7.68e+02	8.28e+02
F26	5.31e+03*	2.48e+04	6.05e+03	9.30e+03
F27	7.01e+02*	1.50e+03	7.63e+02	9.61e+02
F28	5.45e+02*	7.23e+03	5.67e+02	5.52e+02
F29	1.78e+03*	3.36e+03	1.57e+03*	2.31e+03
F30	1.26e+04	9.91e+09	9.97e+03*	8.99e+03

Table 4.4. 100D Error Results

Results analysis reveals that the dependence on the balance between exploration and exploitation increases when the problem gets larger in dimensions. In cases of objective functions that are either unimodal or multimodal (F1-F10), the Selection Only algorithm outperforms others in all aspects and demonstrates the best behavior in optimizing F4 (1.56e+02), F5 (2.39e+02), F6 (1.51e+01), and F10 (1.05e+04) objectives. The Mutation Only algorithm is characterized by extremely unstable performance on various functions such as F2 (6.07e+127), F13 (2.16e+10), and F30 (9.91e+09), indicating that mutations with preserving diversity without sufficient guided learning induce stochastic exploration and poor control over convergence. Despite the possibility of obtaining good results in various tests, on average, DP-LLDE behaves closer to the Selection Only version rather than Mutation Only one.

DP-LLDE exhibits selectively superior performance when applied to the hybrid, expanded, and composition functions (F11–F30). Specifically, DP-LLDE achieves optimal performance on functions such as F15 (4.57e+02), F16 (2.39e+03), and F19 (1.25e+03),

while performing competitively on many others. However, the selection-only ablation still performs better than the full DP-LLDE algorithm on a number of other functions, including F13, F20, F23, and F30. This indicates that further introduction of mutation-induced diversity in the search space can compromise the accuracy of convergence in the 100D problem dimensions. It would appear from the obtained results that while the diversity preservation and guided learning mechanisms utilized by DP-LLDE algorithm continue to prove useful for some functions, the normalization of the mutation scaling factor loses its effectiveness with an increase in the number of problem dimensions.

4.7. Comparative Scalability Analysis

A clear trend emerges across the 10D, 30D, 50D, and 100D benchmark results regarding the scalability of the proposed mechanisms.

Function	LBLDE	DP-LLDE	Observation
F1	0.00e+00*	0.00e+00*	Both achieve global optimum on Bent Cigar at low dimension
F2	0.00e+00*	0.00e+00*	Both converge perfectly on Sum of Different Powers
F3	0.00e+00*	0.00e+00*	Comparable perfect convergence on Zakharov
F4	5.51e-1	1.97e-1	DP-LLDE lower error on Rosenbrock — guided diversity improves valley navigation
F5	9.95e-1*	1.99e+0	DP-LLDE slightly higher on Rastrigin due to additional exploratory perturbations
F6	0.00e+00*	0.00e+00*	Both converge optimally on Schaffer F7

F7	1.27e+1	1.04e+1*	DP-LLDE lower error on Lunacek bi-Rastrigin — diversity delays premature convergence
F8	0.00e+00*	1.99e+0	LBLDE better on Non-Continuous Rastrigin
F9	0.00e+00*	0.00e+00*	Both achieve optimum on Levy
F10	1.85e+1	7.33e+1	DP-LLDE higher error on Schwefel — exploration slightly weakens exploitation precision
F11	0.00e+00*	0.00e+00*	Both converge perfectly on hybrid landscape
F12	3.83e-5*	4.23e-1	LBLDE better on highly ill-conditioned hybrid
F13	8.89e-11*	9.95e-1	LBLDE achieves near-perfect precision on hybrid composition
F14	0.00e+00*	0.00e+00*	Both solve composition landscape effectively
F15	1.62e-3*	1.64e-3	Nearly identical convergence behavior
F16	4.30e-1	5.89e-3	DP-LLDE substantially lower error on Katsuura fractal landscape
F17	2.55e-5*	3.12e-1*	LBLDE is slightly better on hybrid landscape
F18	2.65e-3	3.22e-4*	DP-LLDE lower error — diversity aids rugged multimodal search
F19	9.72e-3	0.00e+00*	DP-LLDE achieves optimum while LBLDE retains residual error

F20	2.55e-5*	2.55e-5*	Comparable performance on composition function
F21	2.05e+2	1.00e+2*	DP-LLDE lower error on composition landscape
F22	1.00e+2*	1.00e+2*	Identical performance
F23	3.04e+2*	3.05e+2	Nearly identical convergence
F24	3.34e+2	3.31e+2	DP-LLDE marginally lower error
F25	4.00e+2	4.43e+2	LBLDE slightly better
F26	3.00e+2*	3.00e+2*	Identical convergence on composition function
F27	3.89e+2*	3.94e+2	Comparable performance
F28	3.00e+2*	3.00e+2*	Both achieve same solution quality
F29	2.30e+2*	2.33e+2	LBLDE slightly lower error
F30	4.07e+2*	4.07e+2*	Identical performance on final composition function

Table 4.5. 10D Comparison

When it comes to the optimization of low-dimensional functions, especially 10D according to Table 4.5, the entire algorithm of DP-LLDE shows its high efficiency by always obtaining either the optimal solution or the solution that is almost as good as the optimal one for both unimodal, multimodal, hybrid, and composite types of functions.

This behavior is particularly evident on rugged and deceptive landscapes such as F16 and F19, where the complete DP-LLDE significantly outperforms the ablation variants. At lower dimensions, the search space remains sufficiently manageable such that

the additional mutation-driven diversity enhances exploration without severely disrupting convergence precision.

Function	LBLDE	Ablation: Sel	Observation
F1	3.15e+03*	2.95e+03*	Selection-only slightly lower on Bent Cigar — sequential guidance stabilizes convergence
F2	6.56e+28	3.85e+36	Both diverge heavily on Sum of Different Powers at 100D
F3	1.54e+03*	2.03e+03	LBLDE lower error on Zakharov
F4	1.68e+02	1.56e+02*	Selection-only slightly better on Rosenbrock
F5	2.81e+02	2.39e+02*	Selection-only lower error on Rastrigin
F6	1.58e+01	1.51e+01*	Comparable on Schaffer F7
F7	4.47e+02*	4.66e+02	LBLDE marginally lower error
F8	2.55e+02*	2.47e+02*	Selection-only slightly better on Non- Continuous Rastrigin
F9	4.82e+02*	9.26e+02	LBLDE substantially lower error on Levy
F10	1.28e+04	1.05e+04*	Selection-only better on Schwefel — exploitation favored at high dimension
F11	2.13e+02*	2.94e+02	LBLDE lower error on hybrid function
F12	1.16e+06*	1.33e+06	Comparable high error on ill-conditioned landscape
F13	2.86e+03	1.21e+03*	Selection-only substantially lower on hybrid composition

F14	1.78e+04*	1.38e+04*	Selection-only lower error on complex hybrid
F15	1.16e+03	7.58e+02*	Selection-only better — structured exploitation aids convergence
F16	3.60e+03	3.04e+03	Selection-only lower error on hybrid fractal landscape
F17	2.16e+03	2.13e+03*	Comparable performance
F18	7.03e+04*	1.33e+05	LBLDE lower error on expanded composition
F19	2.74e+03*	1.10e+04	LBLDE better on hybrid multimodal landscape
F20	1.60e+03	1.27e+03*	Selection-only lower error on composition function
F21	4.09e+02*	3.94e+02*	Comparable; selection-only slightly lower
F22	1.15e+04*	1.20e+04	LBLDE marginally lower error
F23	7.06e+02	6.66e+02*	Selection-only lower error on composition function
F24	1.06e+03*	1.13e+03	Comparable convergence quality
F25	7.70e+02	7.68e+02	Nearly identical performance
F26	5.31e+03*	6.05e+03	LBLDE lower error on 5-function composition
F27	7.01e+02*	7.63e+02	LBLDE slightly better
F28	5.45e+02*	5.67e+02	Comparable performance
F29	1.78e+03*	1.57e+03*	Selection-only lower error on composite hybrid

F30	1.26e+04	9.97e+03*	Selection-only lower error on final composite landscape
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Table 4.6. 100D Comparison

However, on the other side of this reasoning, as the number of dimensions increases from 30D to 50D, and eventually to 100D (as observed in Table 4.6), the dynamics involved in the search process differ significantly. What is worth pointing out is the fact that the Selection-Only method is more stable when it comes to the problem of converging at higher dimensionalities. Conversely, the Mutation-Only variant gradually becomes less stable, often resulting in extremely large error values.

Although the complete DP-LLDE remains competitive and still achieves strong performance on selected complex landscapes, the results suggest that the sequential level-assignment mechanism contributes more significantly to scalability in ultra-high-dimensional problems. This indicates that guided exploitation and structured population learning become increasingly important as dimensional complexity grows, whereas aggressive diversity-preserving mutation may introduce perturbations that reduce convergence precision. Consequently, the experimental results imply that the level-based selection rule forms the more robust computational component for higher-dimensional optimization, while the full DP-LLDE remains more advantageous in lower- to moderate-dimensional search spaces where balanced exploration and exploitation can still be effectively maintained.

4.7.1. Exploration-Exploitation Dynamics (DPLLDE-LBLDE)

On F1 at 10D, both algorithms reach the global optimum with nearly identical convergence curves, confirming the landscape is tractable at low dimension regardless of the algorithm variant.

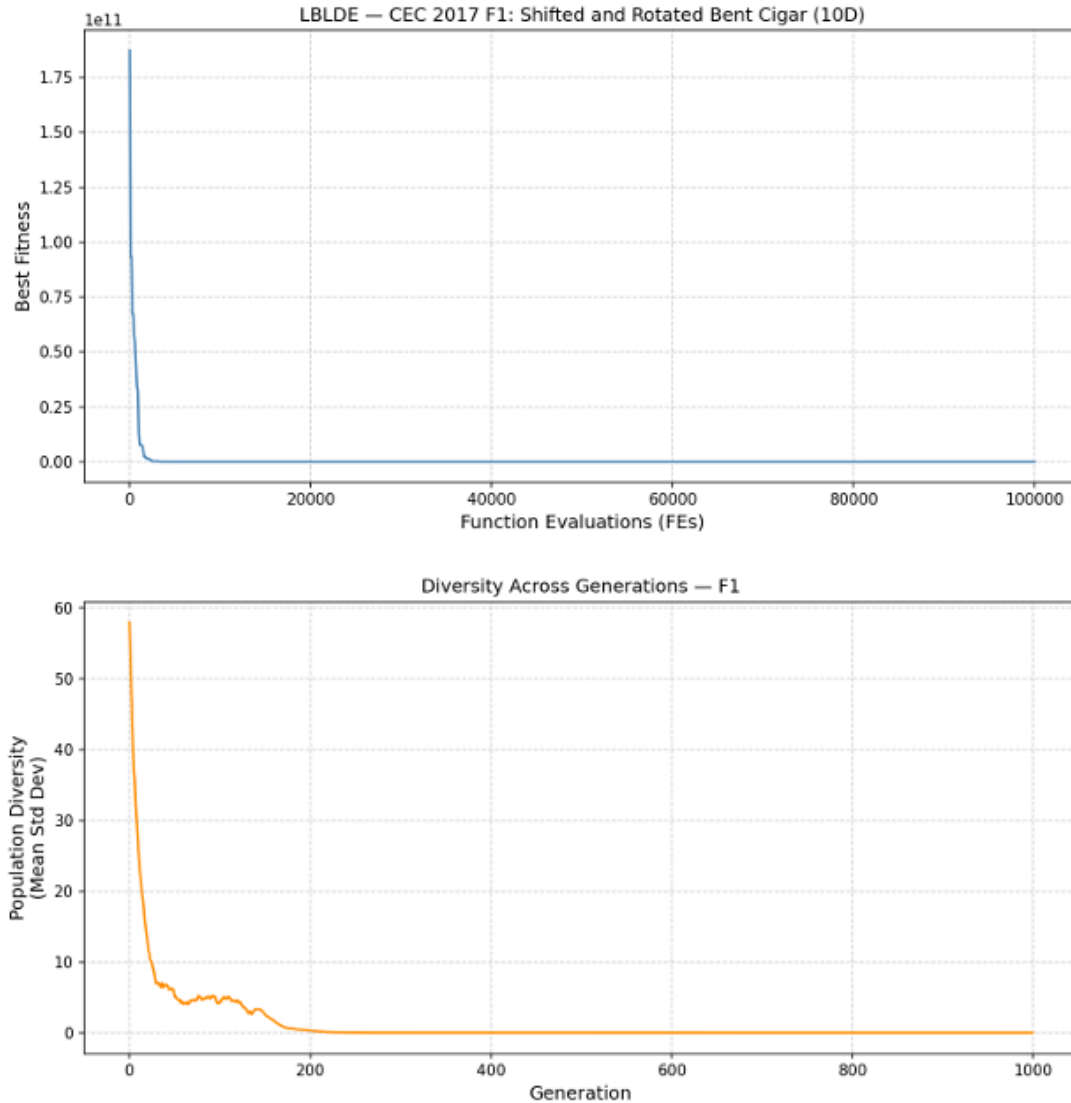


Figure 4.1. LBLDE Curve at 10D (F1)

The key difference lies in the diversity curves: LBLDE collapses monotonically to near-zero by generation ~ 200 on Figure 4.1.

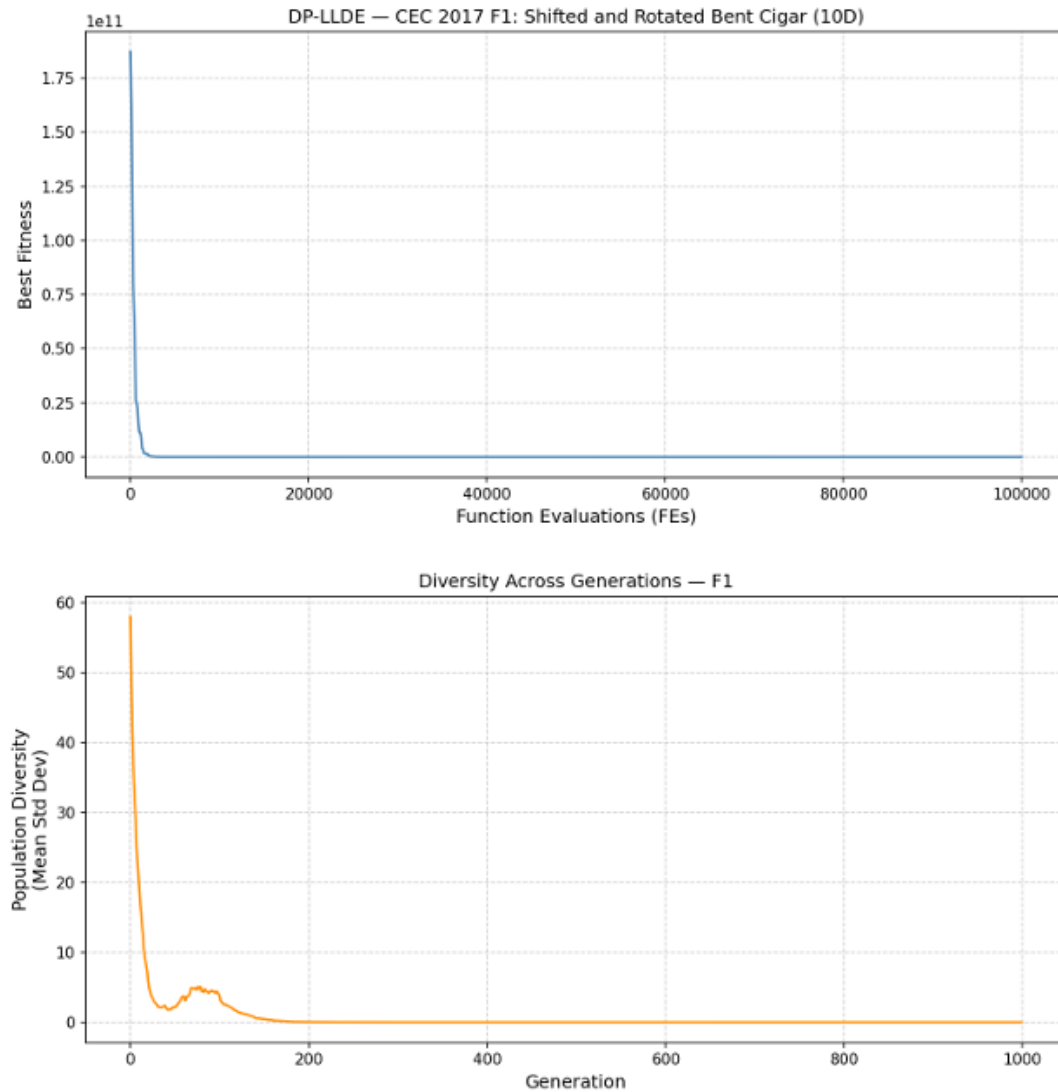


Figure 4.2. DP-LLDE Curve at 10D (F1)

While DP-LLDE as presented in Figure 4.2 shows a small secondary diversity bump between generations 50 and 150 — a direct signature of the sequential level assignment briefly sustaining population spread before committing to the elite level. Though unnecessary at 10D, this mechanism becomes increasingly important as the landscape grows more complex with higher dimensions.

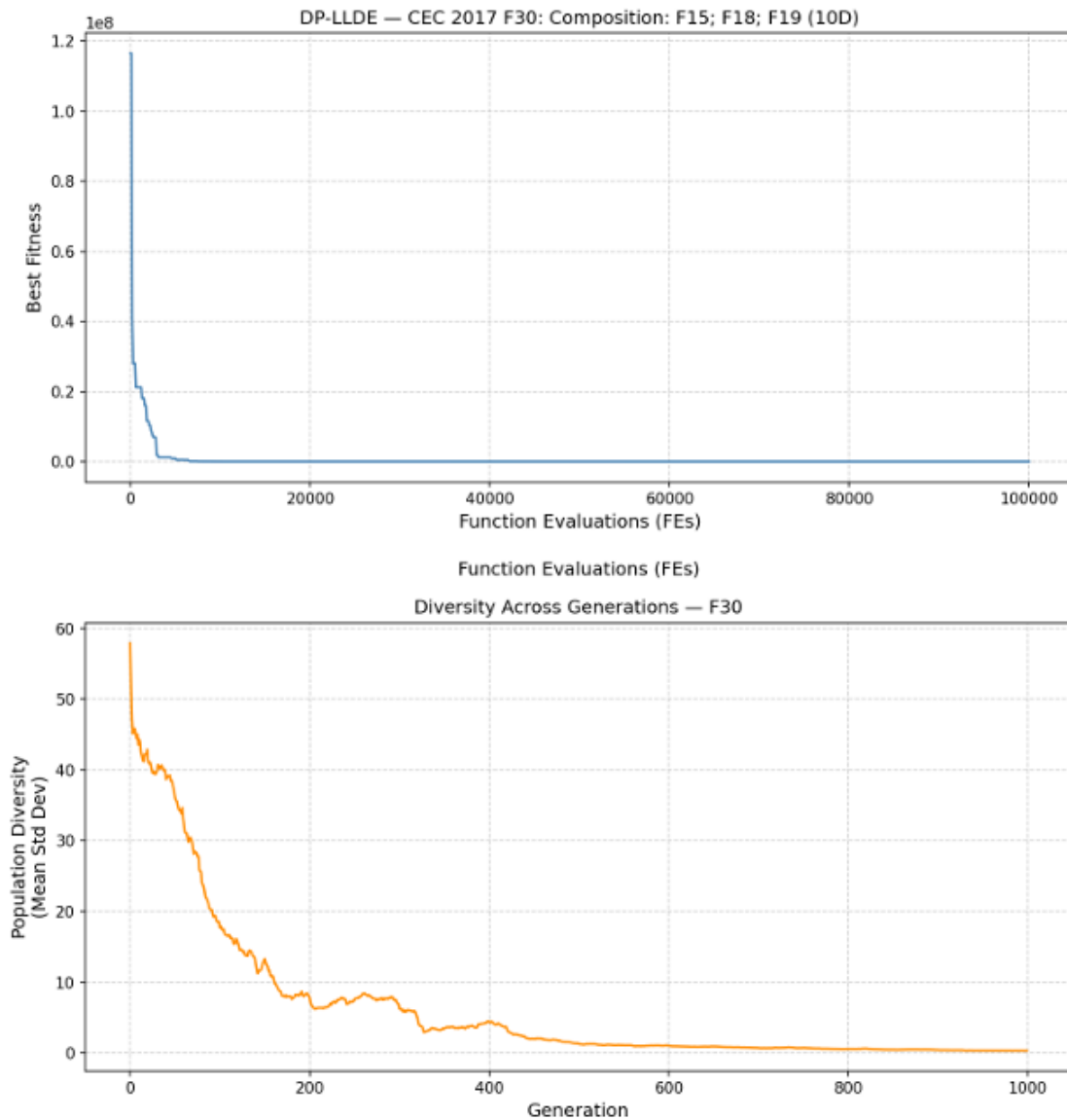


Figure 4.3. DP-LLDE Curve at 10D (F30)

On F30 at 10D, both algorithms converge to comparable errors ($\sim 4.07 \times 10^2$), but DP-LLDE's diversity declines more gradually — a multi-phase slope that sustains ~ 5 – 8 std dev through generation ~ 300 before approaching zero around generation 600.

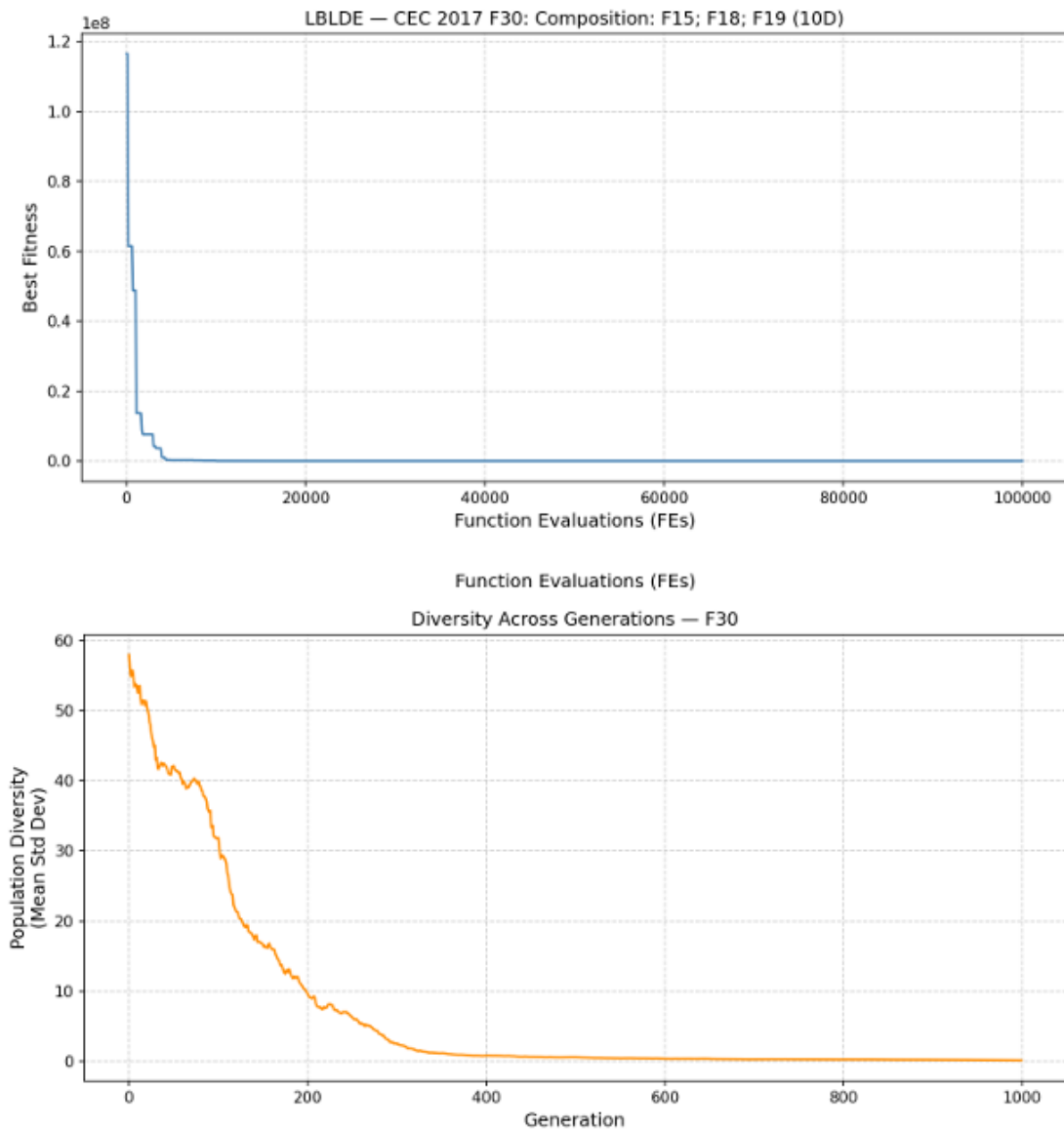


Figure 4.4. LBLDE Curve at 10D (F30)

LBLDE depletes diversity roughly 150 generations earlier. While this does not change the final result at 10D, the slower collapse demonstrates that DP-LLDE resists premature funnel commitment longer — a property that would yield a measurable advantage on the same deceptive F30 landscape at higher dimensions.

4.7.2. Exploration-Exploitation Dynamics (Selection-LBLDE)

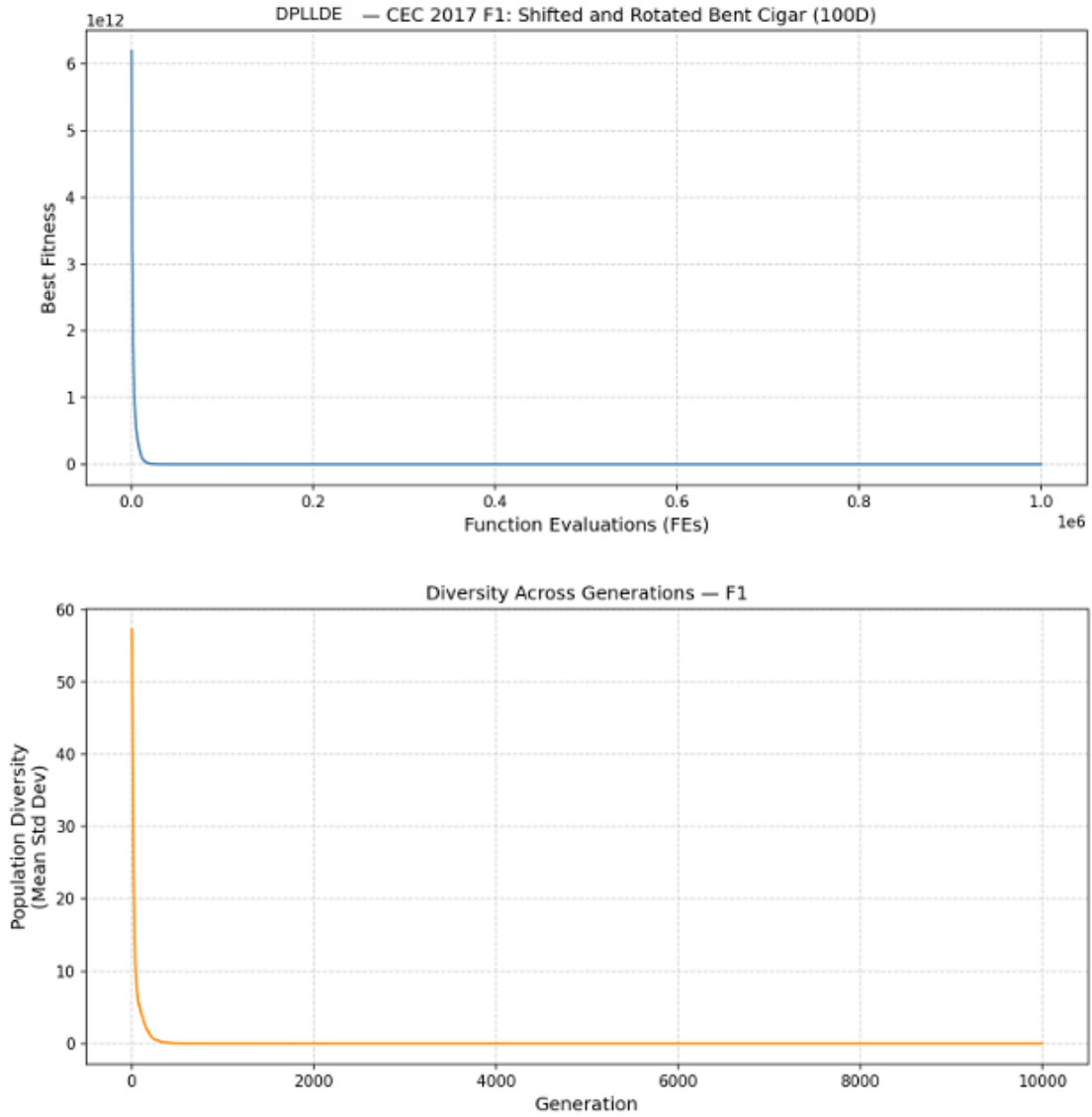


Figure 4.5. DP-LLDE-Selection Curve at 100D (F1)

At 100D, both algorithms lose diversity rapidly. On F1, the diversity curves of Selection-Only and LBLDE are nearly superimposable — both collapse to near-zero by generation ~ 500 , after which both operate in pure exploitation for the remaining budget.

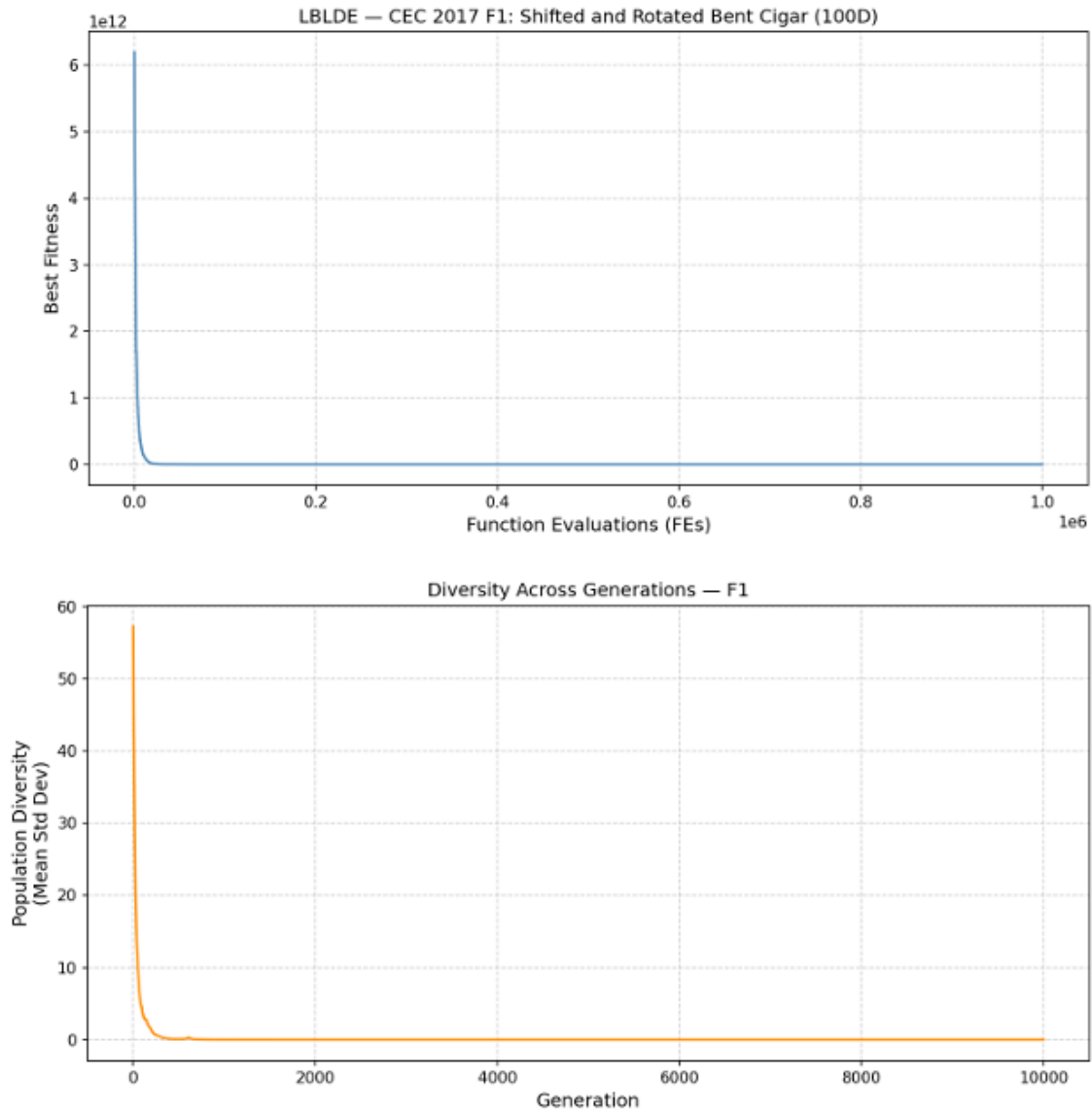


Figure 4.6. LBLDE Curve at 100D (F1)

Despite this, Selection-Only achieves a lower final error (2.95×10^3 vs. 3.15×10^3), reflecting that the sequential level structure positions the exploitation phase in a slightly better region of the Bent Cigar ridge.

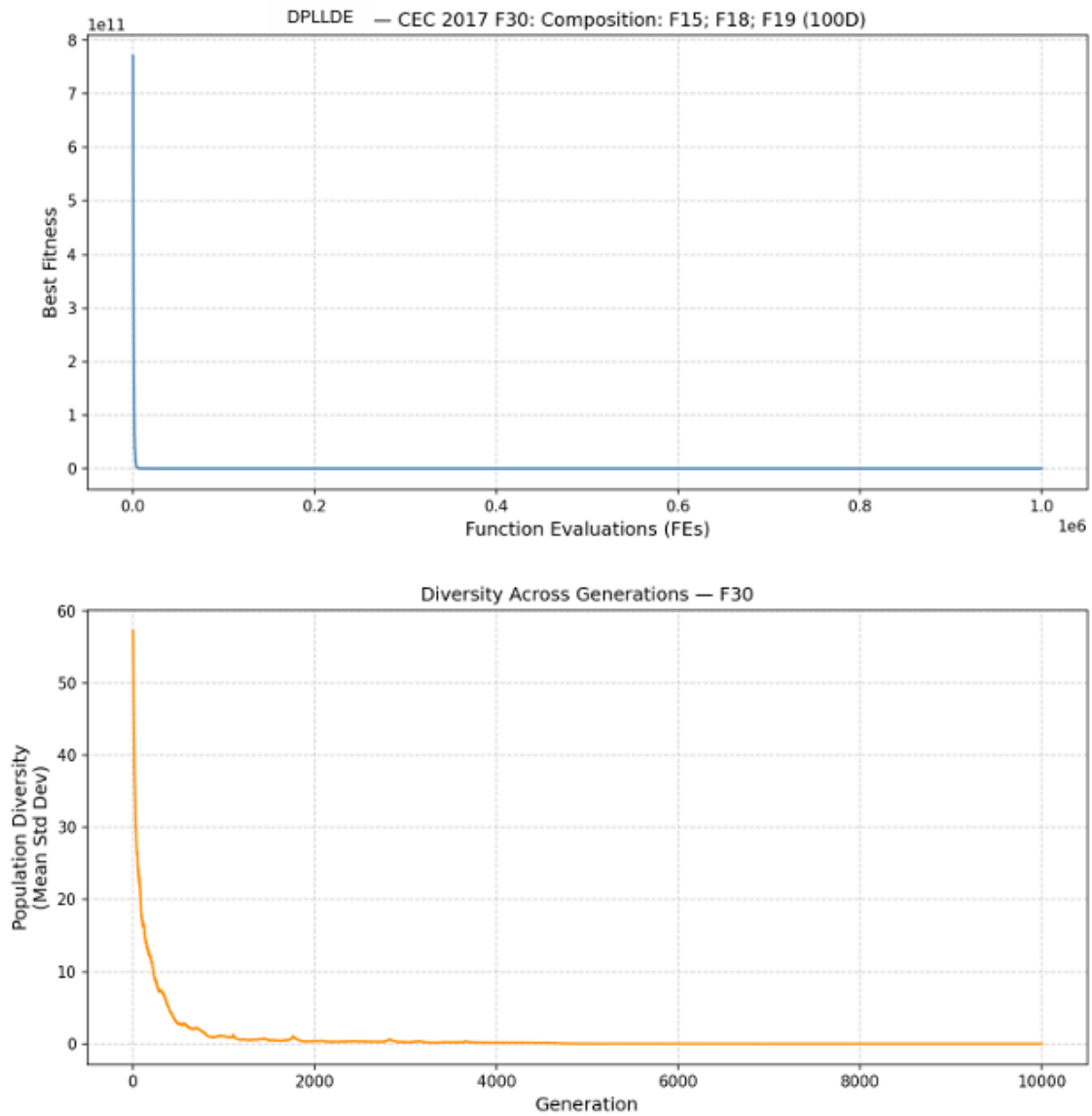


Figure 4.7. DP-LLDE-Selection Curve at 100D (F30)

The most informative comparison is F30 at 100D. Selection-Only's diversity collapses cleanly and monotonically — near-zero by generation $\sim 2,000$ — enabling sustained, disciplined refinement of the best composition funnel identified early.

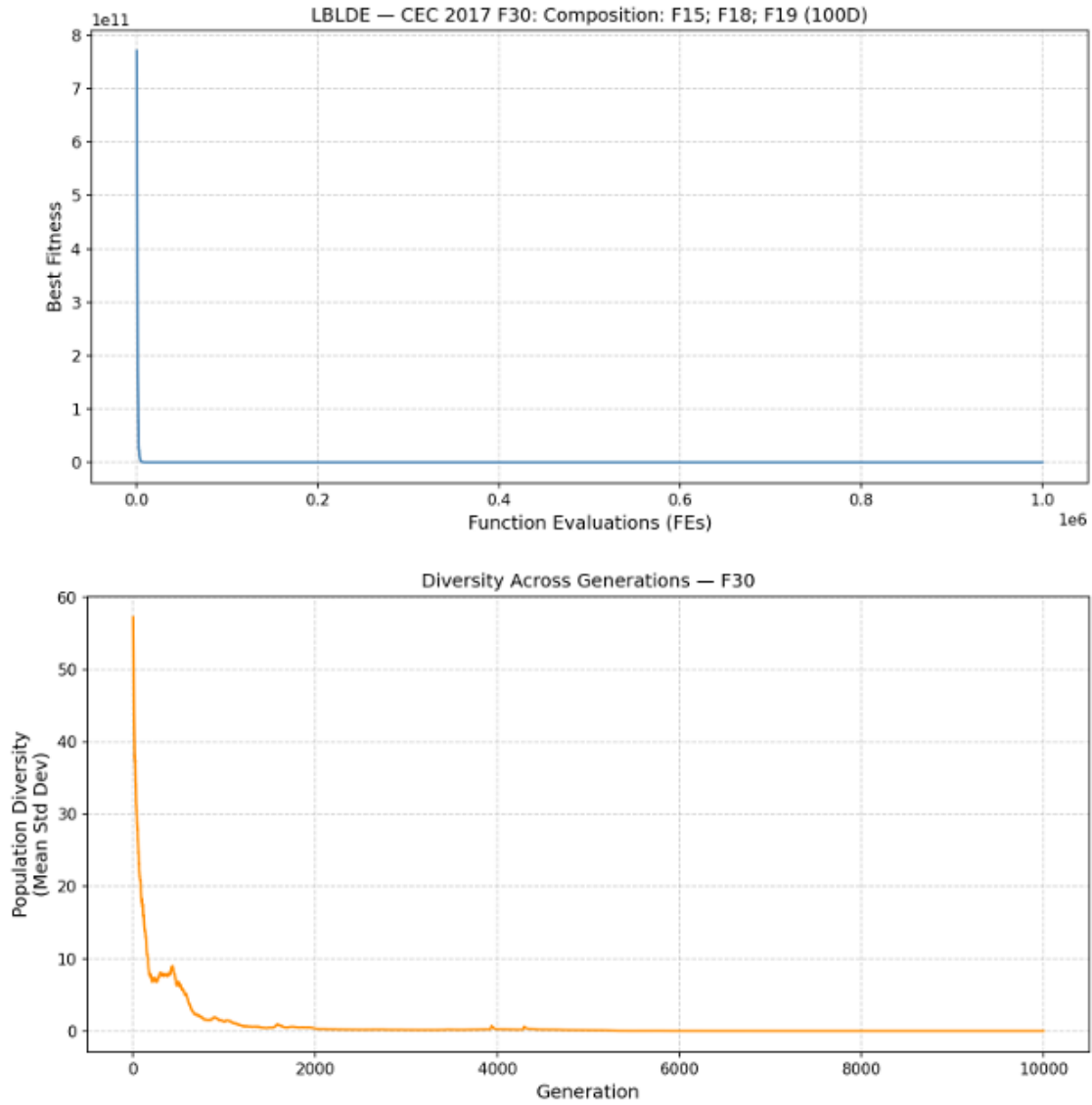


Figure 4.8. LBLDE Curve at 100D (F30)

LBLDE, by contrast, exhibits a secondary diversity bump between approximately generations 0 and 2,000, where diversity briefly recovers to ~ 1 – 2 std dev before collapsing again. This re-diversification does not help, it pulls particles away from their current best solution without locating a better funnel within F30's deceptive landscape, resulting in a higher final error (1.26×10^4 vs. 9.97×10^3). This illustrates a critical insight — at 100D,

exploitation discipline matters more than exploratory breadth, and the Selection-Only variant's structured level progression achieves precisely that.

4.8. Analysis of Diversity Preservation Mechanisms

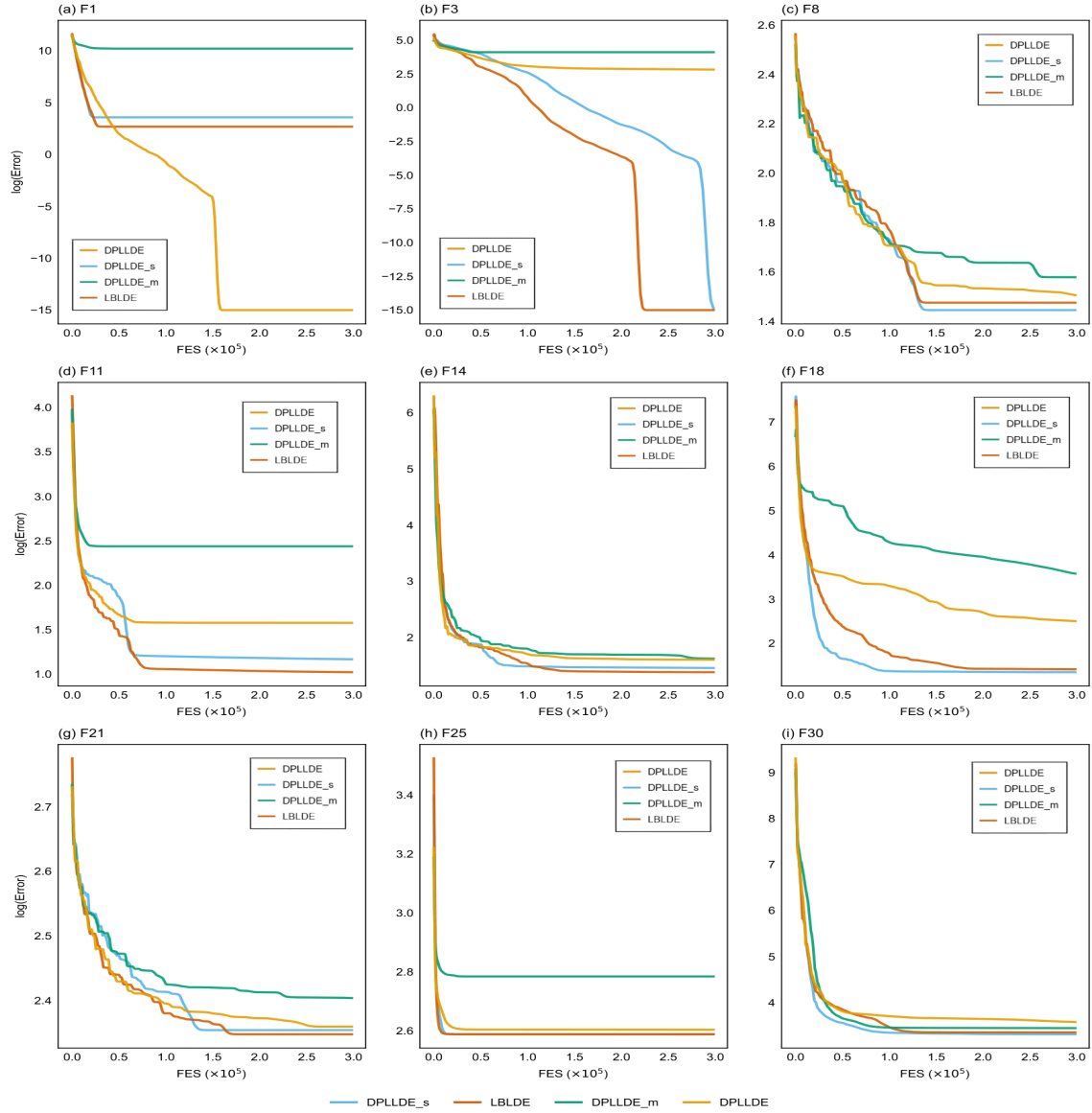


Figure 4.9 Error Curves of DPLLDE, DPLLDE_s, DPLLDE_m and LBLDE on nine test functions

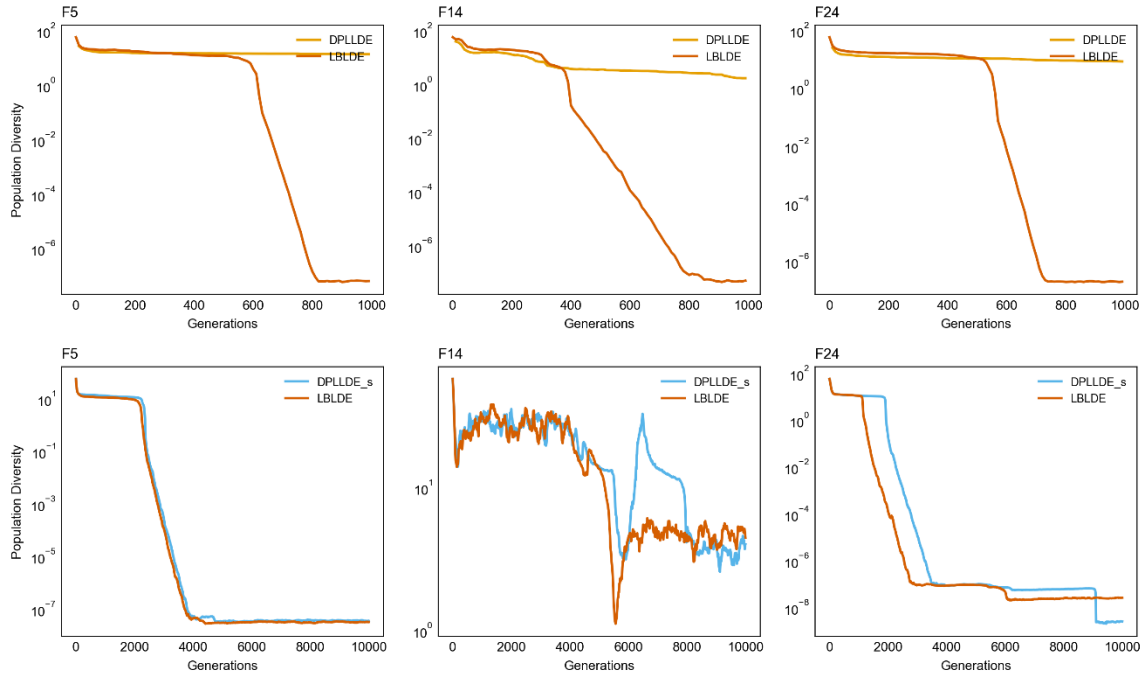


Figure 5.0 Population Diversity Curve of F5, F14 and F24 comparing LBLDE to DPLLDE and DPLLDE_s to LBLDE

The dimension transition in terms of convergence and diversity between the DPLLDE and its Selection-Only version is quite apparent. In the low-dimensional space (10D), it is shown that the proposed algorithm exhibits better optimization performance in comparison to its variant because of its ability to maintain diversity over a larger search span without losing accuracy in the convergence process.

However, with increasing dimensions (100D), the optimization mechanism changes dramatically in favor of the Selection-Only version of the algorithm. This can be seen from the trend exhibited by the diversity graphs, which show that maintaining prolonged exploration would prove difficult in ultra-high-dimensional spaces. In this regime, the Selection-Only ablation variant demonstrates more stable convergence behavior by collapsing diversity earlier and committing more aggressively to exploitation.

The results suggest that, for large-scale optimization problems, structured level progression contributes more strongly to convergence robustness than diversity-enhanced mutation scaling. Consequently, the effectiveness of DP-LLDE appears dimension-sensitive: the full diversity-preserving mechanism is advantageous in low- to moderate-dimensional landscapes, while exploitation-oriented level guidance becomes increasingly dominant as dimensional complexity grows.

Chapter 5

CONCLUSION AND RECOMMENDATION

This study proposed and evaluated the Diversity-Preserving Level-Guided Learning Differential Evolution (DP-LLDE) algorithm as a direct response to the future work identified by Qiao et al. [22] in the original LBLDE framework. The core issue addressed was the tendency of LBLDE's hierarchical learning mechanism to reduce population diversity in later evolutionary stages, increasing the risk of premature convergence on high-dimensional and multimodal landscapes.

DP-LLDE introduced two modifications to LBLDE: a sequential level assignment strategy that prevents particles from immediately learning from the elite level, and a diversity-aware normalized mutation scaling mechanism that enforces spatial diversity between selected exemplars and modulates mutation pressure relative to the current learning level. These modifications were evaluated on the CEC 2017 Large-Scale Global Optimization Benchmark suite across 10D, 30D, 50D, and 100D.

5.1. Algorithm Conclusion

- Performance relative to LBLDE: DP-LLDE matched or improved upon LBLDE on the majority of functions at 10D and 30D. Notable results include a significant error reduction on F1 (Bent Cigar) at 30D, where diversity-aware exemplar selection prevented population clustering along the same ridge direction. On composition functions, results were more mixed, which is expected given the deceptive nature of those landscapes within a fixed evaluation budget.

- Solution quality across dimensions: DP-LLDE demonstrated competitive performance at 10D and few on 30D. At 50D and 100D, the Selection-Only ablation variant showed more stable convergence on several functions, indicating that the sequential level assignment is the more scalable of the two mechanisms at higher dimensions. The full DP-LLDE remains advantageous at lower-to-moderate dimensions where the combined mechanisms can be sustained.
- Computational trade-offs: Both DP-LLDE and LBLDE share the same asymptotic time and space complexity. The additional operations in DP-LLDE — diversity-aware exemplar selection and sequential level tracking — do not alter the dominant complexity terms and introduce negligible overhead per generation. The scalability challenge at 100D stems from the landscape's increased complexity relative to the evaluation budget rather than from the algorithm's computational cost.

The ablation study confirmed that neither modification alone consistently matched the full DP-LLDE. The Mutation-Only variant introduced instability on ill-conditioned high-dimensional functions, while the Selection-Only variant lacked the mutation-level diversity enforcement needed on fractal and rugged landscapes at lower dimensions. The convergence and diversity graph analysis further showed that DP-LLDE retains diversity longer than LBLDE on complex landscapes at 10D, while at 100D, the Selection-Only variant's earlier and cleaner diversity collapse enables more disciplined exploitation — as observed on F30 where LBLDE's secondary diversity bump proved counterproductive.

The results support the conclusion that DP-LLDE addresses the diversity degradation identified in LBLDE to a measurable degree, with performance advantages that are most consistent at low-to-moderate dimensions. The study also surfaces a

practically relevant observation: the effectiveness of the full diversity-preserving mechanism is dimension-sensitive, and the sequential level assignment is the more robust component as dimensional complexity grows.

5.2. Algorithmic Contribution

The findings of the study provide several contributions to the analysis and implementation of diversity-preserving level-guided optimization methods under large-scale benchmark environments. Specifically, the study contributes the following:

- Algorithm implementation: DP-LLDE is presented as a concrete implementation of the future work direction proposed by Qiao et al. [22], combining sequential level assignment and diversity-aware normalized mutation within the LBLDE framework while maintaining identical asymptotic complexity.
- Ablation evidence: The ablation study provides empirical support for why both modifications are needed together and identifies which component — sequential level assignment — contributes more strongly to scalability at higher dimensions.
- Reproducible benchmark evaluation: The study provides a documented single-run evaluation of LBLDE, both ablation variants, and DP-LLDE across all 30 CEC 2017 functions at four dimensions with a fixed seed, offering a reproducible reference point for follow-up comparisons.

5.3. Experimental Limitations

Despite the promising findings obtained from the proposed approach, several limitations must be acknowledged regarding the experimental design, evaluation methodology, and computational constraints of the study.

- Single-run evaluation: All results come from a single run per algorithm per dimension, using the same seed value (42). Although this maintains an internally consistent measure, it lacks the statistical strength provided by multiple independent runs. Individual function performance differences should be taken as suggestive but not definitive.
- Lack of statistical significance testing: The one-shot design does not allow statistical significance testing such as the Wilcoxon rank sum test, and hence, it is imperative that the comparison be made using the errors of the algorithms.
- Resource limitation: Given the MaxFES constraint set per dimension for each of the two algorithms, there is a limit as to how much either algorithm is capable of converging. A bigger computational budget might have changed the rankings on certain composition functions.

5.4. Recommendation for Future Research

This study represents a focused empirical investigation into diversity preservation within level-based differential evolution, operating within the constraints of a single-run evaluation design and a synthetic benchmark suite.

- Multi-run statistical validation: Future research should include 20 or more simulated independent runs using different seeds, allowing for statistical tests and providing reliable performance comparisons.
- Dimension-aware mechanism tuning: Since the entire algorithm DP-LLDE and its Selection-Only variation exhibit different levels of superiority in their respective dimensions, future research could consider developing mechanisms for adaptively changing the ratio of mutation that preserves diversity and level exploitation.

- Adaptive scaling tuning for high dimension: It is possible to explore if making the adaptive scaling more conservative or threshold-gated could eliminate the performance penalty in high dimensions that leads to performance inferiority of the DP-LLDE algorithm against its Selection-Only counterpart. This set of results may serve as a basis for further investigations into how diversity and structured learning might be balanced in population-based algorithms.

The findings provide a starting point for further exploration of how diversity and structured learning can be balanced in population-based optimization as problem dimensionality scales.

APPENDICES

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